STABILITY AND INSTABILITY OF RECURRENCE ASYMPTOTICS FOR DISCRETE MEASURES WITH POWER-LAW ATOMS

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Abstract. We continue the study of the recurrence coefficients a_n, b_n of discrete measures $\sum_k w_k \delta_{k^{-p}}$ with power-law atoms. For weights $w_k = k^{-s}\ell(k)$ with $s > 1$, a previous paper proved the pointwise law $n^p b_n \to c(p)$, $n^p a_n \to 2c(p)$, with rate $O(1/n)$, when ℓ is analytic and slowly varying. Here we show by soft methods that the law holds for every slowly varying ℓ, bounded or not, with a rate controlled by how well ℓ can be smoothed in the logarithmic variable. Slow variation cannot be replaced by boundedness: for every $p > 0$, $s > 1$ and every small $\varepsilon > 0$ there are weights $k^{-s}e^{\varepsilon_k}$ with $|\varepsilon_k| \le \varepsilon$ for which $n^p b_n$ does not converge, and others for which $n^p a_n$ does not converge, although the averaged law of the first paper holds for all of them. For pure powers we then let s vary. A Christoffel identity gives $\sum_{m\le n} \log(b_m(s+p)/b_m(s)) + \frac{p}{2}\log n \to \frac{1}{2}\log(c(p)\zeta(s)/\zeta(s+p))$. A variational argument extends it to $\sum_{m\le n} \log(b_m(s_1)/b_m(s_0)) = -\frac{s_1-s_0}{2}\log n + \frac{1}{2}\log(c(p)^{(s_1-s_0)/p}\zeta(s_0)/\zeta(s_1)) + o(1)$ for all $s_0, s_1 > 1$. The same argument gives the law uniformly on compact sets of exponents (for $p \le 1$, inside an explicit window, and for the diagonal coefficients away from one exponent). At $p = 1$ this recovers the averaged $1/n$ coefficient $-(s-1)/2$ for every $s > 1$, by soft methods from the estimates of the second paper. All proofs rest on the first-variation formula for $\log h_n$ under multiplicative changes of the weights, combined with localisation of the spectral weights.

1. Introduction

1.1. **Setting and results.** For $p > 0$, $s > 1$ and a sequence $\ell\colon \mathbb{N} \to (0,\infty)$ consider the discrete measure

$$\sigma_{p,s,\ell} := \sum_{k\ge 1} k^{-s}\ell(k)\,\delta_{k^{-p}}, \qquad \sigma_{p,s} := \sigma_{p,s,1}. \tag{1.1}$$

Its atoms $x_k = k^{-p}$ accumulate only at 0. Let P_n be its monic orthogonal polynomials and $h_n := \int P_n^2\,\mathrm{d}\sigma$. They satisfy the three-term recurrence

$$xP_n = P_{n+1} + a_n P_n + b_n^2 P_{n-1} \qquad (n \ge 0,\ b_0 := 0,\ b_n > 0), \tag{1.2}$$

so that $b_n^2 = h_n/h_{n-1}$, and a_n, b_n are the entries of the Jacobi matrix of the measure. Put

$$I(p) := \frac{\sqrt{\pi}}{2}\,\frac{\Gamma(\frac{1}{p}+\frac{1}{2})}{\Gamma(\frac{1}{p}+1)}, \qquad c(p) := \frac{1}{4}\Big(\frac{\pi}{2I(p)}\Big)^p, \tag{1.3}$$

so that $c(1) = \frac{1}{2}$ and $c(2) = \pi^2/16$. We say that a measure satisfies the *pointwise law* if

$$n^p b_n \longrightarrow c(p) \qquad \text{and} \qquad n^p a_n \longrightarrow 2c(p) \qquad (n\to\infty). \tag{1.4}$$

The first paper of this series [8] proves (1.4) in a doubly averaged form for all weights between $c_1 k^{-s}$ and a summable sequence; the constant $c(p)$ comes from an explicitly solvable constrained

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equilibrium problem. The second paper [9] proves (1.4) itself, with rate $O(1/n)$, by Riemann–Hilbert analysis, when ℓ extends to a slowly varying analytic function in a sector, and in particular for the pure powers $\sigma_{p,s}$. The present paper uses the second paper only through the numbered statements restated in §2.2, and contains no Riemann–Hilbert analysis. It asks which weights the pointwise law tolerates, and how the coefficients of the pure powers depend on s.

Definition 1.1 (slow variation)**.** A sequence $\ell\colon \mathbb{N} \to (0,\infty)$ is *slowly varying* if, for every $\lambda \geq 1$,

$$\operatorname{osc}_n(\lambda) := \max\bigl\{|\log \ell(k) - \log \ell(n)| : \ k \in \mathbb{N},\ n/\lambda \leq k \leq \lambda n\bigr\} \longrightarrow 0 \qquad (n \to \infty).$$

The case $\lambda = 2$ implies the others (Remark 2.12). No boundedness, monotonicity or measurability is involved.

Theorem 1.2 (every slowly varying factor)**.** *Let $p > 0$, $s > 1$, and let ℓ be slowly varying. Then the recurrence coefficients of $\sigma_{p,s,\ell}$ satisfy $n^p b_n \to c(p)$ and $n^p a_n \to 2c(p)$. More precisely,*

$$|n^p b_n - c(p)| + |n^p a_n - 2c(p)| = O\bigl(1/n + \rho_\ell(\lceil n/M_0 \rceil)\bigr), \tag{1.5}$$

where $\rho_\ell(m) \to 0$ is the error of Gaussian smoothing of $\log \ell$ in the logarithmic variable (Definition 3.1) and $M_0 = M_0(p) \geq 4$ is an integer depending only on p (Lemma 2.18).

For example, $\ell(k) = \exp\sqrt{\log k}$ and $\ell(k) = \exp(\sqrt{\log k}\,\sin\log\log k)$ $(k \geq 3)$ are covered; the second is not eventually comparable with any power of $\log k$. The only analytic input is the case of analytic factors, [9, Theorem 1.2], which we apply to a smoothed factor.

Slow variation cannot be replaced by boundedness. To state this, put $u_{\mathrm{sat}}(p) := (4c(p))^{-1/p}$; we show $u_{\mathrm{sat}}(p) < 1$ in Lemma 2.23.

Theorem 1.3 (bounded rough weights)**.** *Let $p > 0$ and $s > 1$.*

(i) *Let $0 < \tau_1 < \tau_2$ with $\tau_2 > u_{\mathrm{sat}}(p)$; every $\tau_2 \geq 1$ qualifies, for example $[\tau_1, \tau_2] = [0.6, 1.6]$. There is $\gamma_W > 0$, depending only on p, τ_1, τ_2 and given by an integral over the arcsine law (4.1), with the following property. For every $\varepsilon \in (0, \gamma_W/16]$ there are integers $N_1 < N_2 < \cdots$ and numbers $\varepsilon_k \in \{-\varepsilon, 0, \varepsilon\}$, vanishing outside $\bigcup_i [\tau_1 N_i, \tau_2 N_i]$, such that the weights $w_k = k^{-s} e^{\varepsilon_k}$ satisfy*

$$\limsup_{n\to\infty} \log \frac{b_n^2}{b_{n-1}^2} \geq \frac{\varepsilon \gamma_W}{4}.$$

(ii) *Let $\mathfrak{i} \in (0, 1]$ be the number defined in (4.4) and $\gamma_{\mathrm{a}} := c(p)\mathfrak{i}/2$. For every $\varepsilon \in (0, \mathfrak{i}/16]$ there are $\varepsilon_k \in \{-\varepsilon, 0, \varepsilon\}$ such that the weights $w_k = k^{-s} e^{\varepsilon_k}$ satisfy*

$$\limsup_{n\to\infty} \bigl(n^p a_n - 2c(p)\bigr) \geq \frac{\varepsilon \gamma_{\mathrm{a}}}{2}, \qquad \liminf_{n\to\infty} \bigl(n^p a_n - 2c(p)\bigr) \leq -\frac{\varepsilon \gamma_{\mathrm{a}}}{2}.$$

(iii) *For all weights with $e^{-\varepsilon} k^{-s} \leq w_k \leq e^{\varepsilon} k^{-s}$, in particular those of* (i) *and* (ii)*, the Hankel determinants and the coefficients b_n of $\sum_k w_k \delta_{k^{-p}}$ satisfy the averaged law of* [8]*, (1.7) and (1.8) below. In case* (i) *the sequence $n^p b_n$ has no limit in $[0, \infty]$; in case* (ii) *the sequence $n^p a_n$ has no limit.*

Numerically $\mathfrak{i} \approx 0.314$, while the proved constant γ_W is very small (about 10^{-7} for $p = 2$ and $[0.6, 1.6]$); the effect itself is of size about ε (Remark 4.5 and Section 7). The only analytic input of Theorem 1.3 is the pure-power law without rate, [9, Corollary 1.3].

For pure powers we write $a_n(s), b_n(s), h_n(s)$ for the quantities of $\sigma_{p,s}$, and ζ for the Riemann zeta function, so that $\sigma_{p,s}$ has total mass $\zeta(s)$. Some of the statements below use a bound for

the spectral weights of $\sigma_{p,s}$ near the accumulation point of the atoms, which follows from [9, Proposition 8.6] and an elementary lemma (Lemma 2.22) for s in the set

$$\mathcal{S}_p := (1, 1+2p) \quad \text{if } 0 < p \le 1, \qquad \mathcal{S}_p := (1, \infty) \quad \text{if } p > 1. \tag{1.6}$$

Theorem 1.4 (pure powers as s varies)**.** *Let $p > 0$.*

(i) *For every $s > 1$,*

$$\sum_{m=1}^{n} \log \frac{b_m(s+p)}{b_m(s)} + \frac{p}{2} \log n \longrightarrow \frac{1}{2} \log \frac{c(p)\zeta(s)}{\zeta(s+p)}.$$

(ii) *For all $s_0, s_1 > 1$,*

$$\sum_{m=1}^{n} \log \frac{b_m(s_1)}{b_m(s_0)} = -\frac{s_1 - s_0}{2} \log n + \frac{1}{2} \log \frac{c(p)^{(s_1-s_0)/p}\zeta(s_0)}{\zeta(s_1)} + o(1).$$

Equivalently $h_n(s_1)/h_n(s_0) = c(p)^{(s_1-s_0)/p} n^{-(s_1-s_0)}(1+o(1))$, and this holds uniformly for s_0, s_1 in compact subsets of $\mathcal{S}_p$.

(iii) *For every compact $K \subset \mathcal{S}_p$, $\sup_{s\in K} |n^p b_n(s) - c(p)| \to 0$. If moreover $p > 1$, or $0 < p \le 1$ and $1 + p \notin K$, then also $\sup_{s\in K} |n^p a_n(s) - 2c(p)| \to 0$.*

Part (i) uses only the limits of [9, Corollary 1.3]; parts (ii) and (iii) also use [9, Proposition 8.6]. In logarithmic Cesàro mean, part (ii) says that the coefficient of $1/n$ in $n^p b_n(s)/c(p)$ decreases in s with slope exactly $-\frac{1}{2}$ (sum the $1/n$ expansion that a pointwise statement would give). An explicit anchor fixes the coefficient at $p = 1$.

Corollary 1.5 (the averaged $1/n$ coefficient at $p = 1$)**.** *For every $s > 1$ there is $C(s) \in \mathbb{R}$ such that the coefficients of $\sigma_{1,s} = \sum_k k^{-s}\delta_{1/k}$ satisfy*

$$\sum_{j=1}^{n} \log\big(2j\, b_j(s)\big) = -\frac{s-1}{2} \log n + C(s) + o(1).$$

Equivalently, with $D_n(s) := \det\big(\zeta(s+i+j)\big)_{0\le i,j\le n-1}$,

$$\log \frac{D_{n+1}(s)}{D_n(s)} = \log \zeta(s) - 2\log n! - 2n\log 2 - (s-1)\log n + 2C(s) + o(1).$$

For $p = 1$ and $s \in \frac{3}{2} + \mathbb{N}_0$ the first statement is proved without the bound of [9, Proposition 8.6], and so is its analogue for $p = 2$ and $s \in 2 + 2\mathbb{N}_0$ (Corollary 5.5). The pointwise $1/n$ term is known for every $p > 0$: $n^p b_n(s) = c(p)(1 - \frac{s-1}{2n} + O(n^{-1-\min(p,1)}))$ by [9, Theorem 1.4], restated as Theorem 2.6 below. Summing this expansion gives the logarithmic mean up to $o(1)$, but not the additive constant; at $p = 1$ it yields the first statement of Corollary 1.5. The proof given here does not use the pointwise $1/n$ term, and Theorem 1.4(ii) adds the exact differences of the constants.

1.2. **Context.** *Averaged and pointwise laws.* Specialised to the atoms k^{-p}, [8, Theorem 1.1 and Corollary 1.2] state that if $\sum_k w_k < \infty$ and $w_k \ge c_1 k^{-s'}$ for some $c_1 > 0$ and $s' \in \mathbb{R}$, then the Hankel determinants D_n of $\sum_k w_k \delta_{k^{-p}}$ satisfy

$$\log D_n = -pn^2 \log n + \kappa_p n^2 + o(n^2), \qquad \kappa_p := \tfrac{3p}{2} + \log c(p), \tag{1.7}$$

equivalently

$$\frac{2}{n^2} \sum_{j=1}^{n-1} (n-j) \log\big(j^p b_j\big) \longrightarrow \log c(p). \tag{1.8}$$

By Heine's formula, multiplying every weight by a factor in $[e^{-\varepsilon}, e^{\varepsilon}]$ changes D_n by a factor in $[e^{-\varepsilon n}, e^{\varepsilon n}]$, so (1.7) cannot distinguish such weights. Theorems 1.2 and 1.3 show that the pointwise law does distinguish them: it is stable under slowly varying perturbations of any size, and unstable under rough ones, however small. We have not found Theorem 1.3 in the literature.

Spectral weights. For a fixed measure with bounded support, Simon [7, Theorems 3.1 and 3.2] relates the convergence of the Jacobi parameters to weak limits of the measures $p_n^2 \, \mathrm{d}\mu$, by counting lattice paths [7, Proposition 3.3]. Our localisation lemma (Lemma 2.20) is the forward half of this argument at the moving scale n^{-p}: the spectral weights of the pure powers, rescaled by n^p, converge to the arcsine law on $[0, 4c(p)]$.

First variation. Fischer [4, Lemma 4, (20)–(21)] computes the derivatives of the recurrence coefficients with respect to a parameter of the orthogonality functional; for a multiplicative change of the weights these are the formulas of Lemmas 2.13 and 2.14. What we add in §2.5 is a bound on the second derivative with absolute constants, which turns the first-order formulas into estimates for perturbations of fixed size.

The Christoffel transformation. Multiplying $\sigma_{p,s}$ by x gives $\sigma_{p,s+p}$. The Jacobi matrix of $x\sigma$ is obtained by factorising that of σ as LU and reversing the factors [2, §1], and the factors form a chain sequence in the sense of Dai, Piao and Qiao [1, Proposition 2.1]. The second paper uses this to transfer the law with its rate from s to $s+p$ [9, Theorem 3.1]. Here the same factorisation, combined with the product structure of h_n, gives the Cesàro-level identities of Section 5.

Special cases. The moments of $\sigma_{1,s}$ are $\zeta(s+j)$, so Corollary 1.5 concerns Hankel determinants of zeta values, studied by Monien [6] and by Haynes and Zudilin [5]; their leading order is [8, Corollary 1.4]. Our anchor at $p = 1$ is the Tricomi–Carlitz measure, whose orthogonality and recurrence are recorded by Wu, Lin, Xu and Zhao [11, (1.4)–(1.9)]; at $p = 2$ it is the Lommel case [9, Remark 1.6], which rests on the recurrence recorded by Van Assche [10, §3].

1.3. **Method.** All proofs use one device. Along a path of weights $w(t) = v\, e^{t\varphi}$, the first variation of $\log h_n$ is the average of φ against the spectral weights $m_n(k) = w_k(t) p_n(x_k)^2$ of the orthonormal polynomial p_n (Lemma 2.13). Two facts locate these weights: the atoms $k < n/M_0$ are saturated, so that m_n is exponentially small there (Lemma 2.18), and for pure powers m_n lives at indices $k \asymp n$, where its image under $x \mapsto n^p x$ converges to the arcsine law (Lemma 2.20). Each later section is one choice of φ. In Section 3, $\varphi = \log(\ell/\ell_1)$ with ℓ_1 a smoothing of ℓ; it is small where m_n lives, which gives stability. In Section 4, $\varphi = \pm\varepsilon$ with the sign of the second difference $m_n - 2m_{n-1} + m_{n-2}$ on a window of indices of size n; the linear response then does not average out, and a second-order bound shows that the remainder cannot cancel it. In Section 5, $\varphi = -p \log k$ is multiplication by x, which shifts s by p. In Section 6, $\varphi = -(s_1 - s_0)\log k$ with $s_1 - s_0$ small; this gives the derivative in s and a variational two-sided bound for $h_n(s_1)/h_n(s_0)$.

1.4. **Organisation and notation.** Section 2 collects the tools: conventions, the results imported from [8, 9], slow variation, the first variation and its second-order bounds, and localisation. Sections 3, 4, 5 and 6 prove Theorems 1.2, 1.3, 1.4(i) and 1.4(ii)–(iii) with Corollary 1.5. Section 7 illustrates the results numerically, and Section 8 states open problems and conjectures.

Throughout, $p > 0$ is fixed and $c = c(p)$. The atoms are $x_k = k^{-p}$, and x also denotes the sequence (x_k). As in [9], the quotient $a_n/(2b_n)$ pairs the diagonal entry at P_n with the off-diagonal entry b_n between P_{n-1} and P_n, as in (1.2). Constants C, C' are positive and may change from line to line; the letter c is reserved for $c(p)$, and c_1 for the lower-bound constant

of the weights in §1.2. Symbols local to this paper are the smoothing error ρ_ℓ and head margin M_0 (Section 3 and Lemma 2.18), the window constants ι_W, γ_W and the number $\mathfrak{i}$ (Section 4), the arcsine tail T_p (Lemma 2.20), the exponent ϑ_* (Lemma 2.22), and Φ_n, Θ_n (Section 6).

2. First variation and localisation

This section collects the tools used in every later section. After fixing conventions (§2.1) we restate the results imported from [8] and [9] (§2.2); a reader who accepts these statements can read the rest of the paper without the earlier papers. We then prove two facts about slowly varying sequences (§2.3), the first-variation formulas (§2.4) with bounds on their second-order remainders (§2.5), and the two localisation properties of the spectral weights (§2.6). The proofs are short and, apart from the second-order bounds, of a standard kind.

2.1. **Conventions.** Throughout, σ denotes a measure $\sum_{k\ge1} w_k\delta_{x_k}$ with distinct atoms $x_k \in (0,1]$, weights $w_k>0$ and $\|w\|_1 := \sum_k w_k < \infty$, except where a more general positive measure on $[0,\infty)$ is allowed explicitly. Such a σ has infinite support and finite moments, so its monic orthogonal polynomials P_n exist for all $n\ge0$ and satisfy (1.2). The orthonormal polynomials $p_n := P_n/\sqrt{h_n}$ satisfy

$$xp_n = b_{n+1}p_{n+1} + a_np_n + b_np_{n-1} \qquad (n\ge0,\ p_{-1}:=0), \tag{2.1}$$

and therefore $a_n = \int xp_n^2\,\mathrm{d}\sigma$ and $b_n = \int xp_{n-1}p_n\,\mathrm{d}\sigma$. The Hankel determinants are $D_n = \prod_{j<n} h_j$. Multiplying all weights by one positive constant changes none of P_n, p_n, a_n, b_n.

We regard functions on $\mathbb{N}$ as vectors and write $\langle f,g\rangle := \sum_k f(k)g(k)$ whenever the sum converges absolutely, $f(A) := \sum_{k\in A} f(k)$ for $A\subset\mathbb{N}$, and $f|_A := f\mathbf{1}_A$. The *spectral weights* of σ are

$$q_n(k) := w_k^{1/2}p_n(x_k), \qquad m_n(k) := q_n(k)^2, \qquad K_n(k) := \sum_{j<n} m_j(k). \tag{2.2}$$

The vectors $q_0, q_1, \dots$ are orthonormal in $\ell^2(\mathbb{N})$, so each m_n is a probability vector and $m_n = K_{n+1} - K_n$. By (2.1), at every atom,

$$x_kq_n(k) = b_{n+1}q_{n+1}(k) + a_nq_n(k) + b_nq_{n-1}(k), \tag{2.3}$$

so that $a_n = \langle x, m_n\rangle$ and $b_n = \langle xq_{n-1}, q_n\rangle \le \tfrac12\langle x, m_{n-1}+m_n\rangle$. We write P_m for the orthogonal projection of $\ell^2(\mathbb{N})$ onto the span of $q_0,\dots,q_{m-1}$; then $K_m(k)$ is the k-th diagonal entry of P_m.

Christoffel's extremal property. For $y\in\mathbb{R}$ let $\mathsf{K}_m(y) := \sum_{j<m} p_j(y)^2$. If $\deg P < m$, then $P = \sum_{j<m}\langle P,p_j\rangle_\sigma p_j$ with $\langle P,p_j\rangle_\sigma := \int Pp_j\,\mathrm{d}\sigma$, and the Cauchy–Schwarz inequality gives $P(y)^2 \le \mathsf{K}_m(y)\int P^2\,\mathrm{d}\sigma$, with equality for $\langle P,p_j\rangle_\sigma = p_j(y)$. Hence

$$\frac{1}{\mathsf{K}_m(y)} = \min\Big\{\int P^2\,\mathrm{d}\sigma :\ \deg P<m,\ P(y)=1\Big\}. \tag{2.4}$$

Every competitor in (2.4) with $y=x_k$ has $\int P^2\,\mathrm{d}\sigma \ge w_k$, so $K_m(k) = w_k\mathsf{K}_m(x_k) \le 1$, and in particular $m_n(k) \le 1 - K_n(k)$.

2.2. **Results imported from the first two papers.** We restate exactly what is used, with the numbering of [8] and [9]. The atoms $x_k = k^{-p}$ are the case $f(\xi)=\xi^{-p}$, $C=1$ of the atoms $f(k)$ in [8], since then $\xi^{p+1}f'(\xi) = -p$.

Theorem 2.1 ([8, Theorem 1.1 and Corollary 1.2])**.** *Let $w_k\ge0$ satisfy $\sum_k w_k<\infty$ and $w_k \ge c_1k^{-s'}$ for all $k\ge1$, with constants $c_1>0$ and $s'\in\mathbb{R}$. Then the Hankel determinants and the coefficients b_n of $\sum_k w_k\delta_{k^{-p}}$ satisfy (1.7) and (1.8).*

For $\beta > 0$ let arc_β be the arcsine law on $[0,\beta]$, with density $1/(\pi\sqrt{y(\beta-y)})$; it is the image of $\mathrm{d}\theta/\pi$ on $(0,\pi)$ under $\theta \mapsto \frac{\beta}{2}(1+\cos\theta)$, and $\mathrm{arc}_\beta([0,y]) = \frac{2}{\pi}\arcsin\sqrt{y/\beta}$ for $0 \le y \le \beta$.

Proposition 2.2 ([8, Theorem 1.3(i)]). *Let $\mu^* := \int_0^1 \mathrm{arc}_{4c\tau^{-p}}\,\mathrm{d}\tau$, a probability measure on $(0,\infty)$, and let μ_u^* be its image under $x \mapsto x^{-1/p}$. Then μ_u^* has a density ρ^* with $\rho^* = 1$ on $(0, u_{\mathrm{sat}}]$ and $0 < \rho^* < 1$ on $(u_{\mathrm{sat}}, \infty)$, where $u_{\mathrm{sat}} = u_{\mathrm{sat}}(p) = (4c)^{-1/p}$.*

Definition 2.3 ([9, Definition 1.1]). Let $\vartheta > 0$. A sequence $\ell\colon \mathbb{N} \to (0,\infty)$ is *analytic of angle ϑ* if $\ell(k) = \exp\Lambda(\log k)$ for all $k \ge 1$, where, for some $c_0 > 0$, the function Λ is analytic on $S_\vartheta := \{\xi \in \mathbb{C} :\ \mathrm{Re}\,\xi \ge -c_0,\ |\mathrm{Im}\,\xi| \le \vartheta\}$, real on $S_\vartheta \cap \mathbb{R}$, bounded on $S_\vartheta \cap \{\mathrm{Re}\,\xi \le 0\}$, and $\sup\{|\Lambda'(\xi)| :\ \xi \in S_\vartheta,\ \mathrm{Re}\,\xi \ge T\} \to 0$ as $T \to \infty$.

Theorem 2.4 ([9, Theorem 1.2]). *Let $s > 1$, and let ℓ be analytic of angle ϑ for some $\vartheta > 0$. Then the coefficients of $\sigma_{p,s,\ell}$ satisfy $n^p b_n = c(1 + O(1/n))$ and $a_n/(2b_n) = 1 + O(1/n)$; equivalently $|n^p a_n - 2c| + |n^p b_n - c| = O(1/n)$.*

(The two forms are equivalent: from the first, $n^p a_n = 2n^p b_n \cdot a_n/(2b_n) = 2c(1+O(1/n))$; from the second, $a_n/(2b_n) = (2c+O(1/n))/(2c+O(1/n))$.) The case $\ell \equiv 1$ is the following.

Theorem 2.5 ([9, Corollary 1.3]). *For every $s > 1$ the coefficients of $\sigma_{p,s}$ satisfy $n^p b_n = c(1+O(1/n))$ and $a_n/(2b_n) = 1 + O(1/n)$. In particular $n^p b_n \to c$ and $n^p a_n \to 2c$.*

Only the last sentence of Theorem 2.5 is used in Sections 4–6. The next statement is quoted only in remarks.

Theorem 2.6 ([9, Theorem 1.4]). *Let $s > 1$.*

(i) *If $p \ge 1$, then $n^p b_n(s) = c(1 - \frac{s-1}{2n} + O(n^{-2}))$ and $n^p a_n(s) = 2c(1 - \frac{s-1+p}{2n} + O(n^{-2}))$.*
(ii) *If $0 < p < 1$, then, with $\mathsf{k}_p := 2p\,c\,\zeta(-p)$,*

$$n^p b_n(s) = c\Big(1 - \frac{s-1}{2n} + \frac{\mathsf{k}_p}{n^{1+p}} + O\big(n^{-1-2p} + n^{-2}\big)\Big),$$
$$n^p a_n(s) = 2c\Big(1 - \frac{s-1+p}{2n} + \frac{\mathsf{k}_p}{n^{1+p}} + O\big(n^{-1-2p} + n^{-2}\big)\Big).$$

In both cases $a_n(s)/(2b_n(s)) = 1 - p/(2n) + O(n^{-1-\min(2p,1)})$.

The Christoffel transformation is the subject of [9, Section 3]. Let σ be a positive Borel measure on $[0,\infty)$ with infinite support and finite moments. By [9, Lemma 3.2], P_n has n simple zeros in $(0,\infty)$, so $(-1)^n P_n(0) > 0$, and one may define

$$u_n := -\frac{P_{n+1}(0)}{P_n(0)} > 0 \quad (n \ge 0), \qquad l_0 := 0, \qquad l_n := \frac{b_n^2}{u_{n-1}} > 0 \quad (n \ge 1). \tag{2.5}$$

Lemma 2.7 ([9, Lemmas 3.3 and 3.4]). *Let σ be as above, with coefficients a_n, b_n, and let $\tilde a_n, \tilde b_n$ be the coefficients of $x\sigma$, $\mathrm{d}(x\sigma) := x\,\mathrm{d}\sigma(x)$.*

(i) *$a_n = u_n + l_n$ for $n \ge 0$, $b_n^2 = u_{n-1} l_n$ for $n \ge 1$; $x\sigma$ has infinite support, its monic orthogonal polynomials are $(P_{n+1} + u_n P_n)/x$, and*

$$\tilde a_n = u_n + l_{n+1} \quad (n \ge 0), \qquad \tilde b_n^2 = u_n l_n \quad (n \ge 1).$$

(ii) *If $c' > 0$, $n^p a_n \to 2c'$ and $n^p b_n \to c'$, then $n^p u_n \to c'$ and $n^p l_n \to c'$.*

Theorem 2.8 ([9, Theorem 3.1], without rate). *Let σ be as above and $c' > 0$. If $n^p a_n(\sigma) \to 2c'$ and $n^p b_n(\sigma) \to c'$, then the same holds for $x\sigma$. For the atoms k^{-p} one has $x\,\sigma_{p,s,\ell} = \sigma_{p,s+p,\ell}$.*

Proposition 2.9 ([9, Remark 1.6]). *For $n \ge 1$ the coefficients of $\sigma_{2,2}$ are $b_n = \pi^2(4n+1)^{-1}((4n-1)(4n+3))^{-1/2}$ and $a_n = 2\pi^2(4n+1)^{-1}(4n+5)^{-1}$.*

The last import is a bound for the spectral weights of the pure powers at indices beyond Mn, that is, near the accumulation point $x = 0$ of the atoms. It is used only in Section 6. Recall the set $\mathcal{S}_p$ of (1.6).

Proposition 2.10 ([9, Proposition 8.6])**.** *Let $s \in \mathcal{S}_p$ (that is, $1 < s < 1 + 2p$, or $p > 1$ and $s > 1$), and let m_n be the spectral weights of $\sigma_{p,s}$. There are $C, \gamma_* > 0$ and n_0 such that $m_{n'}((Mn, \infty)) \le CM^{-p/2}$ for all $n \ge n_0$, $n' \in \{n-1, n\}$ and $1 \le M \le \gamma_* n^{2/p}$.*

For $p > 1$, [9, Proposition 8.6] also gives a bound for all $M \ge 1$ with any exponent below $\min(p/2, s-1)$. We do not need it: Lemma 2.22 below derives a bound for all $M \ge 1$ from Proposition 2.10 alone, with a smaller exponent, and this suffices for Section 6.

2.3. **Slow variation.** Let ℓ be slowly varying and put $\operatorname{osc}_n^* := \sup_{m \ge n} \operatorname{osc}_m(2)$. Each $\operatorname{osc}_m(2)$ is a maximum of finitely many numbers and $\operatorname{osc}_m(2) \to 0$, so osc_n^* is finite, non-increasing, and tends to 0.

Lemma 2.11 (dyadic chaining and Potter bounds)**.** *Let ℓ be slowly varying.*

(i) *For all $n \ge 1$ and all integers $k \ge n$, $|\log \ell(k) - \log \ell(n)| \le \operatorname{osc}_n^*\big(1 + \log_2(k/n)\big)$.*

(ii) *For every $\eta > 0$ there is $A_\eta < \infty$ with $e^{-A_\eta} k^{-\eta} \le \ell(k) \le e^{A_\eta} k^{\eta}$ for all $k \ge 1$.*

Proof. (i) Let $j := \lfloor \log_2(k/n) \rfloor$, so that $2^j n \le k < 2^{j+1} n$. For $0 \le i < j$ the integer $2^{i+1}n$ lies in $[2^i n/2, 2 \cdot 2^i n]$, so $|\log \ell(2^{i+1}n) - \log \ell(2^i n)| \le \operatorname{osc}_{2^i n}(2) \le \operatorname{osc}_n^*$; also $|\log \ell(k) - \log \ell(2^j n)| \le \operatorname{osc}_n^*$. Summing these $j+1$ steps gives the claim, since $j \le \log_2(k/n)$.

(ii) Choose m with $\operatorname{osc}_m^* \le \eta \log 2$. For $k \ge m$, part (i) gives $|\log \ell(k)| \le |\log \ell(m)| + \operatorname{osc}_m^* + \eta \log k$. Take $A_\eta := \max(|\log \ell(m)| + \operatorname{osc}_m^*, \max_{k<m} |\log \ell(k)|)$. □

Remark 2.12. If $\operatorname{osc}_n(2) \to 0$, then $\operatorname{osc}_n(\lambda) \to 0$ for every $\lambda \ge 1$: Lemma 2.11(i) uses only $\operatorname{osc}(2)$, and for $n \le k \le \lambda n$ it gives $|\log \ell(k) - \log \ell(n)| \le \operatorname{osc}_n^*(1 + \log_2 \lambda)$, while for $n/\lambda \le k < n$, applied with base point k, it gives $|\log \ell(n) - \log \ell(k)| \le \operatorname{osc}_{\lceil n/\lambda \rceil}^*(1 + \log_2 \lambda)$.

By Lemma 2.11(ii) with $\eta < s - 1$, $\sum_k k^{-s} \ell(k) < \infty$ for every $s > 1$, so $\sigma_{p,s,\ell}$ is a measure of the kind fixed in §2.1.

2.4. **First variation.** Fix atoms and weights $v = (v_k)$ as in §2.1, a bounded real sequence φ on $\mathbb{N}$, and put

$$w(t) := v\, e^{t\varphi} \qquad (t \in \mathbb{R}).$$

All quantities of §2.1 computed for $w(t)$ carry the argument t when needed, and a dot denotes $\mathrm{d}/\,\mathrm{d}t$. Formulas of this kind appear in Fischer [4, Lemma 4]; we include short proofs in the form needed here. For $\varphi = x$ they reduce to the Toda equations.

Regularity. The moments $\sum_k v_k e^{t\varphi(k)} x_k^i$ are C^∞ in t, with derivatives computed term by term, since the r-th termwise derivative is bounded by $v_k \|\varphi\|_\infty^r e^{|t|\|\varphi\|_\infty}$. The coefficients of P_n solve a linear system whose determinant is $D_n > 0$, so they are C^∞, and so are h_n, a_n, b_n^2. The vectors $w(t)^{1/2} x^j$ are C^∞ maps into $\ell^2(\mathbb{N})$ (the r-th derivative is $(\varphi/2)^r w(t)^{1/2} x^j$), and $q_0, \dots, q_n$ arise from finitely many of them by the Gram–Schmidt process, so they are C^∞ maps into $\ell^2(\mathbb{N})$ as well.

Lemma 2.13 (first variation)**.** *For $n \ge 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}t} \log h_n = \langle \varphi, m_n \rangle, \tag{2.6}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} \log b_n^2 = \langle \varphi, m_n - m_{n-1} \rangle \qquad (n \ge 1), \tag{2.7}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} a_n = \langle (x - a_n)\varphi, m_n \rangle - 2b_n \langle \varphi, q_{n-1} q_n \rangle. \tag{2.8}$$

The right-hand sides do not change if a constant is added to φ.

Proof. For a polynomial $g(x;t)$ whose coefficients are C^1 in t, $\frac{\mathrm{d}}{\mathrm{d}t}\sum_k w_k(t)g(x_k;t) = \sum_k w_k(t)\partial_t g(x_k;t) + \sum_k \varphi(k)w_k(t)g(x_k;t)$, because the left side is a finite combination of moments. Apply this to $g = P_n^2$: since P_n is monic, $\partial_t P_n$ has degree less than n and is orthogonal to P_n, so $\dot h_n = \sum_k \varphi(k)w_k P_n(x_k)^2 = h_n\langle\varphi, m_n\rangle$, which is (2.6); (2.7) follows from $b_n^2 = h_n/h_{n-1}$. For (2.8), expand $\dot p_n = \sum_{j\le n}\mathsf{c}_j p_j$. Differentiating $\int p_n p_j\,\mathrm{d}\sigma = \delta_{nj}$ gives $\mathsf{c}_j = -\langle\varphi, q_n q_j\rangle$ for $j < n$ (the term $\int p_n\dot p_j\,\mathrm{d}\sigma$ vanishes because $\deg\dot p_j \le j < n$) and $\mathsf{c}_n = -\frac12\langle\varphi, m_n\rangle$. Then

$$\dot a_n = 2\int x p_n\dot p_n\,\mathrm{d}\sigma + \langle x\varphi, m_n\rangle = 2\mathsf{c}_n a_n + 2\mathsf{c}_{n-1}b_n + \langle x\varphi, m_n\rangle,$$

by (2.1), which is (2.8). Finally $\langle 1, m_n - m_{n-1}\rangle = 0$, $\langle x - a_n, m_n\rangle = 0$ and $\langle q_{n-1}, q_n\rangle = 0$. □

Lemma 2.14 (Toda form)**.** *Put $\mathcal{E}_n := b_{n+1}q_n q_{n+1}$ (a vector on $\mathbb{N}$; $\mathcal{E}_{-1} := 0$) and $\Delta\mathcal{E}_n := \mathcal{E}_n - \mathcal{E}_{n-1}$. Then at every atom*

$$\Delta\mathcal{E}_n = (x - a_n)m_n - 2b_n q_{n-1}q_n = 2\mathcal{E}_n - (x - a_n)m_n, \tag{2.9}$$

and $\dot a_n = \langle\varphi, \Delta\mathcal{E}_n\rangle$. Moreover $|\Delta\mathcal{E}_n| \le \frac12\max(b_n, b_{n+1})(m_{n-1} + 2m_n + m_{n+1})$ at every atom.

Proof. Multiplying (2.3) by $q_n(k)$ gives $(x_k - a_n)m_n(k) = \mathcal{E}_n(k) + \mathcal{E}_{n-1}(k)$, which yields both forms of (2.9); pairing the first with φ and comparing with (2.8) gives $\dot a_n = \langle\varphi, \Delta\mathcal{E}_n\rangle$. The bound follows from $|q_j q_{j+1}| \le \frac12(m_j + m_{j+1})$. □

Lemma 2.15 (derivative of the spectral vectors)**.** *For $m \ge 0$, with $\alpha_m := \langle\varphi, m_m\rangle$,*

$$\dot q_m = \tfrac12\varphi q_m - \mathsf{P}_m(\varphi q_m) - \tfrac12\alpha_m q_m.$$

Proof. Since $q_m = w(t)^{1/2}p_m(x)$ and $\deg\dot p_m \le m$, we have $\dot q_m = \frac12\varphi q_m + r$ with r in the span of $q_0, \dots, q_m$. For $j < m$ the same reasoning gives $\dot q_j - \frac12\varphi q_j \perp q_m$, so differentiating $\langle q_m, q_j\rangle = 0$ gives $\langle r, q_j\rangle = -\langle\varphi q_m, q_j\rangle$. Differentiating $\langle q_m, q_m\rangle = 1$ gives $\langle r, q_m\rangle = -\frac12\alpha_m$. Hence $r = -\mathsf{P}_m(\varphi q_m) - \frac12\alpha_m q_m$. □

2.5. **Second-order bounds.** The first-variation formulas are exact derivatives; to use them for a perturbation e^φ of fixed size one needs the second derivative. The next lemma bounds it with absolute constants, for every base measure. Because of this uniformity, the constructions of Section 4 need no information about the perturbed measures.

For a bounded real φ put $\varepsilon := \frac12(\sup\varphi - \inf\varphi)$. Write $\Delta^2 m_N := m_N - 2m_{N-1} + m_{N-2}$ and $F_N := \log(b_N^2/b_{N-1}^2)$ for $N \ge 2$. A superscript v denotes quantities of the base weights v, and $w := w(1) = ve^\varphi$.

Lemma 2.16 (linear response with quadratic remainder)**.** *In the setting of §2.4, with $\varepsilon = \frac12\operatorname{osc}\varphi$:*

(i) $|\log b_n^2(w) - \log b_n^2(v) - \langle\varphi, m_n^v - m_{n-1}^v\rangle| \le \varepsilon^2$ *for $n \ge 1$;*
(ii) $|F_N(w) - F_N(v) - \langle\varphi, \Delta^2 m_N^v\rangle| \le 2\varepsilon^2$ *for $N \ge 2$;*
(iii) $|a_n(w) - a_n(v) - \langle\varphi, \Delta\mathcal{E}_n^v\rangle| \le (e^\varepsilon - 1 - \varepsilon)\big(b_n(v) + b_{n+1}(v)\big)$ *for $n \ge 0$.*

Proof. Reduction. With $\bar c := \frac12(\sup\varphi + \inf\varphi)$ we have $ve^{t\varphi} = e^{t\bar c}ve^{t(\varphi-\bar c)}$, and constant factors change none of the quantities involved; the linear terms do not change either (Lemma 2.13). So we may assume $|\varphi| \le \varepsilon$. Since $w(t+h) = w(t)e^{h\varphi}$ and $w(t)$ is again admissible, every bound below proved at $t = 0$ for an arbitrary base holds at every t. We use Taylor's formula $g(1) - g(0) - \dot g(0) = \int_0^1 (1-t)\ddot g(t)\,\mathrm{d}t$.

Second derivative of $\log h_m$. By Lemma 2.13 and Lemma 2.15,

$$\dot\alpha_m = 2\langle \varphi q_m, \dot q_m\rangle = \|\varphi q_m\|^2 - \alpha_m^2 - 2\|\mathsf{P}_m(\varphi q_m)\|^2,$$

since P_m is an orthogonal projection. Because q_m is orthogonal to the range of P_m, the vectors $\mathsf{P}_m(\varphi q_m)$, $\alpha_m q_m$ and the rest of φq_m are pairwise orthogonal, and Bessel's inequality gives $\|\mathsf{P}_m(\varphi q_m)\|^2 \le \|\varphi q_m\|^2 - \alpha_m^2 = \mathrm{Var}_{m_m}(\varphi)$. Hence

$$|\dot\alpha_m| \le \mathrm{Var}_{m_m}(\varphi) \le \sum_k \varphi(k)^2 m_m(k) \le \varepsilon^2.$$

(i) and (ii). By (2.7), $\frac{\mathrm{d}^2}{\mathrm{d}t^2}\log b_n^2 = \dot\alpha_n - \dot\alpha_{n-1}$, of absolute value at most $2\varepsilon^2$, and $\ddot F_N = \dot\alpha_N - 2\dot\alpha_{N-1} + \dot\alpha_{N-2}$, of absolute value at most $4\varepsilon^2$. Taylor's formula gives (i) and (ii), since $\int_0^1 (1-t)\,\mathrm{d}t = \frac12$.

(iii). Put $\gamma_m := \langle \varphi q_m, q_{m+1}\rangle$. By (2.3), $\langle (x-a_n)\varphi, m_n\rangle = b_{n+1}\gamma_n + b_n\gamma_{n-1}$, so (2.8) reads $\dot a_n = b_{n+1}\gamma_n - b_n\gamma_{n-1}$. By (2.7), $\dot b_m = \frac12(\alpha_m - \alpha_{m-1})b_m$; in particular $|\dot b_m| \le \varepsilon b_m$, so $b_m(t) \le e^{\varepsilon t} b_m(0)$. Put $u := \varphi q_m$, $u' := \varphi q_{m+1}$, $\mathsf{P} := I - \mathsf{P}_{m+2}$, $S_m := \langle \mathsf{P}_m u, \mathsf{P}_m u'\rangle$ and $R_m := \langle \mathsf{P}u, \mathsf{P}u'\rangle$. Decomposing along the range of P_m, q_m, q_{m+1} and the range of P (pairwise orthogonal), and using Lemma 2.15 twice, one finds

$$\langle \dot q_m, u'\rangle = \tfrac12\langle u, u'\rangle - S_m - \tfrac12\alpha_m\gamma_m, \qquad \langle u, \dot q_{m+1}\rangle = \tfrac12\langle u, u'\rangle - S_m - \alpha_m\gamma_m - \tfrac12\alpha_{m+1}\gamma_m,$$

and $\langle u, u'\rangle = S_m + \alpha_m\gamma_m + \gamma_m\alpha_{m+1} + R_m$. Adding, $\dot\gamma_m = R_m - S_m + \frac12(\alpha_{m+1} - \alpha_m)\gamma_m$, and therefore

$$\ddot a_n = b_{n+1}\Gamma_n - b_n\Gamma_{n-1}, \qquad \Gamma_m := R_m - S_m + (\alpha_{m+1} - \alpha_m)\gamma_m.$$

We claim $|\Gamma_m| \le \varepsilon^2$. By the Cauchy–Schwarz inequality, first in ℓ^2 and then in $\mathbb{R}^2$, $|R_m - S_m| \le (\|\mathsf{P}u\|^2 + \|\mathsf{P}_m u\|^2)^{1/2}(\|\mathsf{P}u'\|^2 + \|\mathsf{P}_m u'\|^2)^{1/2}$. By the orthogonal decomposition, $\|\mathsf{P}u\|^2 + \|\mathsf{P}_m u\|^2 = \|u\|^2 - \alpha_m^2 - \gamma_m^2 \le \varepsilon^2 - \alpha_m^2 - \gamma_m^2$, and likewise for u' with α_{m+1}. By the inequality between the geometric and arithmetic means, $|R_m - S_m| \le \varepsilon^2 - \gamma_m^2 - \frac12(\alpha_m^2 + \alpha_{m+1}^2)$, while $|(\alpha_{m+1} - \alpha_m)\gamma_m| \le \gamma_m^2 + \frac14(|\alpha_m| + |\alpha_{m+1}|)^2 \le \gamma_m^2 + \frac12(\alpha_m^2 + \alpha_{m+1}^2)$. Adding proves the claim. Hence $|\ddot a_n(t)| \le \varepsilon^2(b_n(t) + b_{n+1}(t)) \le \varepsilon^2 e^{\varepsilon t}(b_n + b_{n+1})(v)$, and Taylor's formula with $\dot a_n(0) = \langle \varphi, \Delta\mathcal{E}_n^v\rangle$ (Lemma 2.14) gives (iii), because $\int_0^1 (1-t)\varepsilon^2 e^{\varepsilon t}\,\mathrm{d}t = e^\varepsilon - 1 - \varepsilon$. □

Remark 2.17. The constants in Lemma 2.16 do not depend on n, on the atoms, or on v. The pointwise bound $|\ddot a_n| \le \varepsilon^2(b_n + b_{n+1})$ is attained: for two atoms ±1, $n = 0$, weights proportional to $(e^\theta, e^{-\theta})$ and $\varphi = \pm\varepsilon$ on the two atoms, $a_0 = \tanh\theta$ and $b_1 = 1/\cosh\theta$, and $|\ddot a_0| = 2\varepsilon^2|\tanh\theta|/\cosh^2\theta$ equals $\varepsilon^2(b_0 + b_1)$ when $\sinh\theta = \pm1$. (This example has atoms ±1; the proof of Lemma 2.16 uses only that the atoms are bounded.)

2.6. **Localisation.** Two facts locate the spectral weights. The first holds for all weights and says that the atoms x_k with k much smaller than n are almost fully occupied by $p_0, \dots, p_{n-1}$. For $p > 0$ and $0 < y < 1$ let

(2.10)

$$g_p(y) := (p-y)\log\frac{1}{y} + (1-y)\log(1-y) - p + (p-1)y, \qquad g_p^+ := \begin{cases} g_p, & p \ge 1, \\ g_p - \log\frac1p, & 0 < p < 1. \end{cases}$$

For $y < p/2$ the first term is at least $\frac{p}{2}\log\frac1y$ and the others are bounded, so $g_p^+(y) \to \infty$ as $y \to 0^+$. The *head margin* $M_0 = M_0(p)$ is the least integer $M_0 \ge 4$ such that $g_p^+ \ge 1$ on $(0, 2/M_0]$.

Lemma 2.18 (saturation of the head)**.** *Let $\sigma = \sum_k w_k \delta_{k^{-p}}$ with $w_k > 0$ and $\|w\|_1 < \infty$. For integers $1 \le k < n$ and $y := k/n$,*

$$m_n(k) \le 1 - K_n(k) \le \frac{\|w\|_1}{w_k} \Big(\frac{n+1}{k}\Big)^{2(1-p)_+} e^{-2ng_p^+(y)}.$$

Proof. The first inequality was noted after (2.4). Use the test polynomial $P(x) := \prod_{j\le n,\, j\ne k}(x - x_j)/(x_k - x_j)$ in (2.4): it has degree $n-1$, $P(x_k) = 1$, and it vanishes at x_j for $j \le n$, $j \ne k$. For $i > n$ and $j \le n$ we have $0 < x_i < x_j$, so $|P(x_i)| \le \mathcal{P} := \prod_{j\le n,\, j\ne k} x_j/|x_k - x_j|$. Hence $1/\mathsf{K}_n(x_k) \le w_k + \|w\|_1 \mathcal{P}^2$, and therefore $1 - K_n(k) \le \|w\|_1 \mathcal{P}^2/w_k$. To bound $\mathcal{P}$, note $x_j/|x_k - x_j| = k^p/|k^p - j^p|$ and $|k^p - j^p| \ge \varsigma_p \max(j,k)^{p-1}|k-j|$ with $\varsigma_p := \min(1,p)$ (directly for $p \ge 1$, by the mean value theorem for $p < 1$). Thus

$$\mathcal{P} \le \varsigma_p^{-(n-1)} \frac{k^{k-1}}{(k-1)!} \cdot \frac{k^{p(n-k)} (n!/k!)^{1-p}}{(n-k)!}.$$

Use $k^{k-1}/(k-1)! \le e^k$, $\log(n-k)! \ge (n-k)\log(n-k) - (n-k)$, and $I_{k,n} \le \log(n!/k!) \le I_{k,n} + \log\frac{n+1}{k}$ with $I_{k,n} := \int_k^n \log t \, dt$, taking the lower bound when $p \ge 1$ and the upper bound when $p < 1$. With $k = yn$ the terms in $n \log n$ cancel, and $k + p(n-k)\log k + (1-p)I_{k,n} - (n-k)\log(n-k) + (n-k) = -ng_p(y)$. Hence $\log \mathcal{P} \le -ng_p(y) + (n-1)\log(1/\varsigma_p) + (1-p)_+ \log\frac{n+1}{k}$, and squaring gives the claim. □

Corollary 2.19 (head mass)**.** *Let w be as in Lemma 2.18 with $w_k \ge c_1 k^{-A}$ for all k, where $c_1 > 0$ and $A \ge 0$. For every integer $m \ge 2$,*

$$m_m\big([1, 2m/M_0]\big) \le \frac{\|w\|_1}{c_1} \, m^{A+1}(m+1)^2 e^{-2m}.$$

Proof. For $1 \le k \le 2m/M_0$ we have $k \le m/2 < m$ and $g_p^+(k/m) \ge 1$, and $((m+1)/k)^{2(1-p)_+} \le (m+1)^2$. Apply Lemma 2.18 and sum over fewer than m indices. □

The second fact concerns the pure powers $\sigma_{p,s}$ and is where Theorem 2.5 enters. Put

$$T_p(M) := \mathrm{arc}_{4c}\big([0, M^{-p})\big) = \frac{2}{\pi} \arcsin \sqrt{\min\big(1, M^{-p}/(4c)\big)} \qquad (M \ge 1), \tag{2.11}$$

which decreases to 0 as $M \to \infty$.

Lemma 2.20 (arcsine localisation)**.** *Let $s > 1$, let m_n be the spectral weights of $\sigma_{p,s}$, and fix $i \in \mathbb{Z}$. Let $\nu_{n,i}$ be the image of m_{n+i} under $k \mapsto n^p x_k$, a probability measure on $(0,\infty)$. Then $\nu_{n,i} \to \mathrm{arc}_{4c}$ weakly as $n \to \infty$. In particular $m_{n+i}((Mn,\infty)) \to T_p(M)$ for every $M \ge 1$.*

Proof. Let $\mathcal{J}$ be the Jacobi matrix of $\sigma_{p,s}$, with diagonal $(a_0, a_1, \dots)$ and off-diagonal $(b_1, b_2, \dots)$. By (2.3), $\int y^j \, d\nu_{n,i}(y) = n^{pj} \langle x^j q_{n+i}, q_{n+i}\rangle = n^{pj} (\mathcal{J}^j)_{n+i,n+i}$, which is a sum over lattice paths of length j from $n+i$ to itself, with steps $0, \pm 1$, of products of entries a_m, b_m with $|m - n| \le j + |i| + 1$; compare [7, Proposition 3.3]. For $n > j + |i| + 1$ the set of paths does not depend on n, and by Theorem 2.5 $n^p a_m \to 2c$ and $n^p b_m \to c$ for each such m, since $(m/n)^p \to 1$. So the moment converges to the corresponding sum for the doubly infinite matrix J_0 with diagonal $2c$ and off-diagonal c, that is, to the constant Fourier coefficient of $(c(2 + e^{i\theta} + e^{-i\theta}))^j$, which equals $\frac{1}{\pi}\int_0^\pi (2c(1+\cos\theta))^j \, d\theta = \int y^j \, d\,\mathrm{arc}_{4c}$.

Moment convergence implies weak convergence here, because the limit has compact support. The second moments are bounded, so $(\nu_{n,i})_n$ is tight; let ν' be a subsequential weak limit. Since all moments of $\nu_{n,i}$ are bounded, each y^j is uniformly integrable, so $\int y^j \, d\nu' = \int y^j \, d\,\mathrm{arc}_{4c} \le (4c)^j$. Then $\nu'((4c+\delta,\infty)) \le (4c/(4c+\delta))^{2j} \to 0$ for every $\delta > 0$, so ν' lives on $[0, 4c]$, where

polynomials are dense in the continuous functions; hence $\nu' = \mathrm{arc}_{4c}$. Finally, $m_{n+i}((Mn,\infty)) = \nu_{n,i}([0,M^{-p}))$, and arc_{4c} has no atoms. □

Beyond the range of Proposition 2.10, a soft bound suffices. It uses two elementary facts about a real polynomial P of degree at most n, with $\|P\| := \max_{[-1,1]}|P|$ and T_j the Chebyshev polynomials, $T_j(\cos\theta) = \cos j\theta$:

$$\max_{[-1,1]}|P'| \le 2n^3\|P\|, \qquad |P(y)| \le 2(n+1)\,|T_n(y)|\,\|P\| \quad (|y|\ge 1). \tag{2.12}$$

Indeed, $P = \sum_{j\le n} c_j T_j$ with $c_j = \frac{2-\delta_{j0}}{\pi}\int_0^\pi P(\cos\theta)\cos j\theta\,\mathrm{d}\theta$, so $|c_j| \le 2\|P\|$. On $[-1,1]$, $|T_j'(\cos\theta)| = |j\sin j\theta/\sin\theta| \le j^2$, since $|\sin j\theta| \le j|\sin\theta|$; this gives the first bound. For $|y|\ge 1$, $|T_j(y)| = \cosh(j\,\mathrm{arccosh}\,|y|)$ is non-decreasing in j, which gives the second.

Lemma 2.21 (polynomial tail localisation)**.** *Let $\sigma = \sum_k w_k\delta_{k^{-p}}$ with $c'k^{-s} \le w_k \le C'k^{-s}$ for all k, where $s>1$ and $0<c'\le C'$. For $n\ge 2$ put $T' := \lceil 16pn^3\rceil$ and $T'' := \lceil T'n^{2/p}\rceil$. Then $K_{n+1}(k) \le 68^2(n+1)^2(C'/c')(T''/k)^s$ for every $k \ge T''$, and for every integer $T \ge T''$ and $n' \in \{n-1,n\}$,*

$$m_{n'}\big((T,\infty)\big) \le \frac{68^2(n+1)^2(C'/c')}{s-1}\,T''^{\,s}\,T^{1-s}.$$

Proof. Put $\bar x := x_{T'}$ and $\underline{x} := x_{T''}$, so that $\underline{x}/\bar x \le (T'/T'')^p \le n^{-2} \le \frac14$. Let $\deg P \le n$ and write $\|P\|_{[\underline{x},\bar x]}$ for the maximum of $|P|$ on $[\underline{x},\bar x]$. Mapping $[\underline{x},\bar x]$ affinely onto $[-1,1]$, (2.12) gives $|P'| \le 4n^3\|P\|_{[\underline{x},\bar x]}/(\bar x-\underline{x}) \le 8n^3\|P\|_{[\underline{x},\bar x]}/\bar x$ on $[\underline{x},\bar x]$. Consecutive atoms in $[\underline{x},\bar x]$ are at distance $x_j - x_{j+1} \le pj^{-p-1} \le p\bar x/T'$, so every point of $[\underline{x},\bar x]$ is within $p\bar x/T'$ of an atom, and $\|P\|_{[\underline{x},\bar x]} \le \max|P(x_j)| + (8pn^3/T')\|P\|_{[\underline{x},\bar x]}$, the maximum over $T'\le j\le T''$. Since $T' \ge 16pn^3$, $\|P\|_{[\underline{x},\bar x]} \le 2\max_{T'\le j\le T''}|P(x_j)|$. For $0\le x\le \underline{x}$ the affine image $y = (\underline{x}+\bar x-2x)/(\bar x-\underline{x})$ satisfies $1\le y\le 1+4\underline{x}/\bar x \le 1+4n^{-2}$, and $T_n(1+\eta) = \cosh(n\,\mathrm{arccosh}(1+\eta)) \le e^{n\sqrt{2\eta}} \le e^{\sqrt8} < 17$ for $\eta \le 4n^{-2}$. By (2.12), $|P(x)| \le 34(n+1)\|P\|_{[\underline{x},\bar x]}$. Hence, if $k\ge T''$ and $P(x_k)=1$, then $1 \le 68(n+1)\max_{T'\le j\le T''}|P(x_j)|$, so $\int P^2\,\mathrm{d}\sigma \ge \min_{j\le T''} w_j/(68(n+1))^2 \ge c'T''^{-s}/(68(n+1))^2$. By (2.4), $K_{n+1}(k) = w_k\mathsf{K}_{n+1}(x_k) \le 68^2(n+1)^2(C'/c')(T''/k)^s$. Finally $m_{n'} \le K_{n'+1} \le K_{n+1}$ for $n'\le n$, and $\sum_{k>T}k^{-s} \le T^{1-s}/(s-1)$. □

Lemma 2.22 (one power for all M)**.** *Let $s\in\mathcal{S}_p$, let m_n be the spectral weights of $\sigma_{p,s}$, and put*

$$\vartheta_* = \vartheta_*(p,s) := \frac{s-1}{4+(2+2/p)s}.$$

Then $0<\vartheta_*<\min(p/2,s-1)$, ϑ_* increases with s, and there are C and n_0 such that $m_{n'}((Mn,\infty)) \le CM^{-\vartheta_*}$ for all $n\ge n_0$, $n'\in\{n-1,n\}$ and $M\ge 1$.

Proof. Put $B := 3+(2+2/p)s$, so that $\vartheta_* = (s-1)/(B+1)$, and $\gamma_1 := 1/\vartheta_* = (B+1)/(s-1)$; then $\gamma_1 > 2+2/p$, and $\vartheta_* < (s-1)p/(2s) < p/2$. The map $M\mapsto m_{n'}((Mn,\infty))$ is non-increasing. For $1\le M\le \gamma_* n^{2/p}$ use Proposition 2.10 and $M^{-p/2}\le M^{-\vartheta_*}$. For $\gamma_* n^{2/p} \le M \le n^{\gamma_1}$, monotonicity and Proposition 2.10 give the bound $C\gamma_*^{-p/2}n^{-1} \le C\gamma_*^{-p/2}M^{-\vartheta_*}$, since $M^{\vartheta_*}\le n$. For $M\ge n^{\gamma_1}$ apply Lemma 2.21 with $c'=C'=1$ and $T := \lfloor Mn\rfloor \ge Mn/2$; since $T'' \le (16p+2)n^{3+2/p}$ and $\gamma_1+1 > 3+2/p$, we have $T\ge T''$ for large n, and $\{k>Mn\}\subset\{k>T\}$. This gives $m_{n'}((Mn,\infty)) \le Cn^2n^{(3+2/p)s}(Mn)^{1-s} = Cn^BM^{-(s-1)}$, and $n^BM^{-(s-1)} = n^BM^{-B\vartheta_*}M^{-\vartheta_*} \le M^{-\vartheta_*}$ because $M^{\vartheta_*}\ge n$. □

Lemma 2.23. *$u_{\mathrm{sat}}(p) < 1$ for every $p>0$.*

Proof. By Proposition 2.2, $\mu_u^*((0,u_{\mathrm{sat}}]) = u_{\mathrm{sat}}$. In the variable $x = u^{-p}$ the set $u \le u_{\mathrm{sat}}$ is $x \ge 4c$, so $u_{\mathrm{sat}} = 1-\mu^*((0,4c))$. For $\tau \in [\frac12, 1]$ the law $\mathrm{arc}_{4c\tau^{-p}}$ gives the mass $\frac{2}{\pi}\arcsin(\tau^{p/2}) \ge \frac{2}{\pi}\arcsin(2^{-p/2})$ to $[0,4c)$, so $\mu^*((0,4c)) > 0$. □

3. Every slowly varying factor

This section proves Theorem 1.2. A slowly varying sequence is close to an analytic one: Gaussian smoothing of $\log \ell$ in the logarithmic variable produces an entire function Λ_ℓ whose derivatives tend to 0 in every horizontal strip and which differs from $\log \ell$ by $o(1)$ at the integers (§3.1). The factor $\ell_1 := e^{\Lambda_\ell(\log\cdot)}$ is therefore analytic of every angle, and Theorem 2.4 applies to it. It remains to compare the weights $k^{-s}\ell(k)$ and $k^{-s}\ell_1(k)$, whose ratio tends to 1, along the path of §2.4 with $\varphi = \log(\ell/\ell_1)$ (§3.2). The first variation is an average of φ against spectral weights that live at indices $k \ge n/M_0$, up to an exponentially small head, and there $|\varphi|$ is at most the smoothing error.

One cannot compare ℓ directly with the constant 1: ℓ may be unbounded and need not be close to 1 anywhere, and controlling $\log \ell(k) - \log \ell(n)$ against the spectral weights far beyond $k \asymp n$ would require information about the accumulation point of the atoms. Comparing ℓ with its own smoothing avoids this.

3.1. Gaussian smoothing.

Definition 3.1 (smoothing)**.** Let $L_\ell\colon \mathbb{R} \to \mathbb{R}$ be the piecewise-linear interpolant of the points $(\log k, \log \ell(k))$, $k \ge 1$, extended by the constant $\log \ell(1)$ on $(-\infty, 0]$. Let $G(y) := e^{-y^2/2}/\sqrt{2\pi}$ and

$$\Lambda_\ell := L_\ell * G, \qquad \rho_\ell(m) := \sup_{k \ge m} \bigl|\log \ell(k) - \Lambda_\ell(\log k)\bigr| \qquad (m \ge 1).$$

The function G extends to an entire function, and for $\xi = T + i\upsilon$ with $T, \upsilon, y \in \mathbb{R}$,

$$|G(\xi - y)| = G(T-y)\, e^{\upsilon^2/2}. \tag{3.1}$$

Lemma 3.2 (Gaussian smoothing)**.** *Let ℓ be slowly varying and write $L = L_\ell$, $\Lambda = \Lambda_\ell$.*

- (o) *For every $\eta > 0$, $|L(y)| \le A_\eta + \eta(|y| + \log 2)$ for all $y \in \mathbb{R}$, with A_η as in Lemma 2.11(ii). The function Λ is entire and real on $\mathbb{R}$.*
- (i) *$\rho_\ell(m) \to 0$ as $m \to \infty$.*
- (ii) *For every $\vartheta \ge 0$, as $\operatorname{Re}\xi \to +\infty$, $\Lambda(\xi) - \Lambda(\operatorname{Re}\xi) \to 0$ and $\Lambda^{(j)}(\xi) \to 0$ for every $j \ge 1$, uniformly in $|\operatorname{Im}\xi| \le \vartheta$; and Λ is bounded on $\{\operatorname{Re}\xi \le 0,\ |\operatorname{Im}\xi| \le \vartheta\}$.*
- (iii) *Consequently $\ell_1(k) := e^{\Lambda(\log k)}$ is analytic of angle ϑ for every $\vartheta > 0$ (Definition 2.3, with any $c_0 > 0$), and $\psi := \ell/\ell_1$ satisfies $\sup_{k\ge m} |\log \psi(k)| = \rho_\ell(m)$.*

The proof shows that L has small oscillation on unit intervals far to the right, chains this into a bound for $|L(y) - L(T)|$ that grows linearly in $|y - T|$, and integrates the bound against the derivatives of G, which decay like a Gaussian even at complex arguments by (3.1). We include the short proof, which uses only the sequence form of slow variation.

Proof. (o). For $y \ge 0$ choose $k \ge 1$ with $y \in [\log k, \log(k+1)]$. Then $L(y)$ is a convex combination of $\log \ell(k)$ and $\log \ell(k+1)$, each of absolute value at most $A_\eta + \eta \log(k+1) \le A_\eta + \eta(y + \log 2)$ by Lemma 2.11(ii). For $y < 0$, $|L(y)| = |\log \ell(1)| \le A_\eta$. Since L grows at most linearly, the integrand $L(y)G(\xi - y)$ is dominated, for ξ in a bounded set, by an integrable function of y, and it is entire in ξ; so Λ is entire and may be differentiated under the integral sign. It is real on $\mathbb{R}$ because L and G are.

Local oscillation of L. For $T \ge T_0 := 1 + \log 2$ put $\operatorname{osc}_L(T) := \sup\{|L(y) - L(y')| :\ y, y' \in [T-1, T+1]\}$ and $\operatorname{osc}_L^*(T) := \sup_{T' \ge T} \operatorname{osc}_L(T')$. If $y \in [T-1, T+1] \cap [\log k, \log(k+1)]$, then

$k \ge e^{T-1}/2$ and $k+1 \le e^{T+1}+1$. With $m := \lceil e^T \rceil$ and $\lambda := 2e^2$ we get $k, k+1 \in [m/\lambda, \lambda m]$, so $\operatorname{osc}_L(T) \le 2\operatorname{osc}_m(2e^2)$. Hence $\operatorname{osc}_L(T) \to 0$, and $\operatorname{osc}_L^*(T)$ is finite and tends to 0.

Chaining. Let $T \ge 2T_0$. If $y \ge T$, join T to y by $\lfloor y-T \rfloor$ steps of length 1 and one shorter step; every step lies in an interval $[T'-1, T'+1]$ with $T' \ge T$, so $|L(y)-L(T)| \le \operatorname{osc}_L^*(T)(1+|y-T|)$. If $T/2 \le y < T$, the same argument started at $y \ge T_0$ gives $|L(y)-L(T)| \le \operatorname{osc}_L^*(T/2)(1+|y-T|)$. If $y < T/2$, then $|y| + T \le 5|y-T|$, and (o) with $\eta = 1$ gives $|L(y)-L(T)| \le C_1(1+|y-T|)$ with $C_1 := \max(2A_1+2, 5)$. In all cases

$$|L(y)-L(T)| \le \big(\operatorname{osc}_L^*(T/2)\,\mathbf{1}_{\{y \ge T/2\}} + C_1 \mathbf{1}_{\{y<T/2\}}\big)(1+|y-T|). \tag{3.2}$$

(ii). The derivatives of G are $G^{(j)} = \mathsf{h}_j G$ with polynomials h_j of degree j; let $\bar{\mathsf{h}}_j$ have the absolute values of the coefficients of h_j as coefficients, so that $|\mathsf{h}_j(t+i\upsilon)| \le \bar{\mathsf{h}}_j(|t|+\vartheta)$ for $|\upsilon| \le \vartheta$. The entire function $\xi \mapsto \int_{\mathbb{R}} G(\xi - y)\,\mathrm{d}y$ equals 1 on $\mathbb{R}$, hence everywhere, and therefore $\int_{\mathbb{R}} G^{(j)}(\xi-y)\,\mathrm{d}y = \delta_{j0}$ for all ξ. For $\xi = T + i\upsilon$ with $|\upsilon| \le \vartheta$ and $T \ge 2T_0$, this, (3.1), (3.2) and the substitution $t = T - y$ give

$$\big|\Lambda^{(j)}(\xi) - \delta_{j0}L(T)\big| = \Big|\int_{\mathbb{R}} \big(L(y)-L(T)\big)G^{(j)}(\xi-y)\,\mathrm{d}y\Big| \le e^{\vartheta^2/2}\Big[\operatorname{osc}_L^*(T/2)\int_{\mathbb{R}} \mathsf{g}_j + C_1 \int_{|t|>T/2} \mathsf{g}_j\Big],$$

where $\mathsf{g}_j(t) := \bar{\mathsf{h}}_j(|t|+\vartheta)(1+|t|)G(t)$ is integrable. The right side tends to 0 as $T \to \infty$, uniformly in $|\upsilon| \le \vartheta$. For $j = 0$, applying this at ξ and at $\operatorname{Re}\xi$ and subtracting gives $\Lambda(\xi) - \Lambda(\operatorname{Re}\xi) \to 0$; for $j \ge 1$ it gives $\Lambda^{(j)}(\xi) \to 0$. For $\operatorname{Re}\xi = T \le 0$: $L(y) = L(0)$ for $y \le 0$ and $|L(y) - L(0)| \le 2A_1 + 2\log 2 + y$ for $y > 0$, while $G(T-y) \le G(y)$ for $y > 0$; hence $|\Lambda(\xi) - L(0)| \le e^{\vartheta^2/2}\int_0^\infty (2A_1+2+y)G(y)\,\mathrm{d}y$.

(i). Since $L(\log k) = \log \ell(k)$, the case $j = 0$, $\upsilon = 0$ of the estimate in (ii) gives $\Lambda(\log k) - \log \ell(k) \to 0$.

(iii). Λ is entire, real on $\mathbb{R}$, bounded on $\{-c_0 \le \operatorname{Re}\xi \le 0, |\operatorname{Im}\xi| \le \vartheta\}$, and $\sup\{|\Lambda'(\xi)| : \operatorname{Re}\xi \ge T, |\operatorname{Im}\xi| \le \vartheta\} \to 0$, by (ii). Finally $\log \psi(k) = \log \ell(k) - \Lambda(\log k)$. □

3.2. **Stability under a factor tending to one.** Fix $p > 0$, $s > 1$ and a slowly varying ℓ, and let Λ_ℓ, ℓ_1, $\psi = \ell/\ell_1$ be as in Lemma 3.2. Put

$$\bar\rho := \sup_k |\log \psi(k)| = \rho_\ell(1) < \infty, \qquad w_k^{(t)} := k^{-s}\ell_1(k)\psi(k)^t \quad (t \in [0,1]),$$

so that $w^{(0)}$ and $w^{(1)}$ are the weights of σ_{p,s,ℓ_1} and $\sigma_{p,s,\ell}$. Since $\ell_1\psi^t = \ell\psi^{t-1}$, we have $e^{-\bar\rho}\ell(k) \le \ell_1(k)\psi(k)^t \le e^{\bar\rho}\ell(k)$. Write $m_n^{(t)}$ for the spectral weights of $w^{(t)}$, call the indices $k < n/M_0$ the *head*, and let

$$\mathfrak{h}_n := \sup_{t\in[0,1]} \max_{n' \in \{n-1, n\}} m_{n'}^{(t)}\big([1, n/M_0)\big).$$

Lemma 3.3 (a priori size)**.** *There are C and n_0, depending on p, s, ℓ, such that $\mathfrak{h}_n \le Cn^{s+5}e^{-2n}$ for all $n \ge 2$, and $a_n(w^{(t)}) \le 2M_0^p n^{-p}$ and $b_n(w^{(t)}) \le 2M_0^p n^{-p}$ for all $n \ge n_0$ and $t \in [0,1]$.*

Proof. By Lemma 2.11(ii) with $\eta = 1$, $\|w^{(t)}\|_1 \le e^{\bar\rho}\sigma_{p,s,\ell}(\mathbb{R})$ and $w_k^{(t)} \ge e^{-\bar\rho - A_1}k^{-s-1}$. For $n \ge 2$ and $n' \in \{n-1, n\}$, the head $[1, n/M_0)$ is contained in $[1, 2n'/M_0]$, so Corollary 2.19 with $A = s+1$ bounds $m_{n'}^{(t)}$ of the head by $Cn^{s+2}(n+1)^2e^{-2(n-1)}$, which gives the first claim. For the second, $a_{n'} = \langle x, m_{n'}\rangle$ and $b_n \le \frac{1}{2}\langle x, m_{n-1} + m_n\rangle$ by §2.1. Off the head $x_k \le (M_0/n)^p$, and $x_k \le 1$ always, so a_n and b_n are at most $(M_0/n)^p + \mathfrak{h}_n$, which is at most $2M_0^p n^{-p}$ for large n. □

Proposition 3.4 (stability). *There are C and n_0, depending on p, s, ℓ, such that for $n \ge n_0$*

$$\big|\log b_n^2(\sigma_{p,s,\ell}) - \log b_n^2(\sigma_{p,s,\ell_1})\big| \le 2\rho_\ell(\lceil n/M_0\rceil) + Cn^{s+5}e^{-2n},$$
$$n^p\big|a_n(\sigma_{p,s,\ell}) - a_n(\sigma_{p,s,\ell_1})\big| \le 6M_0^p\,\rho_\ell(\lceil n/M_0\rceil) + Cn^{p+s+5}e^{-2n}.$$

Proof. Apply §2.4 with $v_k = k^{-s}\ell_1(k)$ and $\varphi = \log\psi$, which is bounded by $\bar\rho$; then $w(t) = w^{(t)}$. Write $\rho := \rho_\ell(\lceil n/M_0\rceil)$. For $k \ge n/M_0$ we have $k \ge \lceil n/M_0\rceil$, so $|\varphi(k)| \le \rho$; on the head $|\varphi(k)| \le \bar\rho$. By (2.7),

$$\big|\log b_n^2(w^{(1)}) - \log b_n^2(w^{(0)})\big| \le \sup_{t\in[0,1]} \sum_k |\varphi(k)|\big(m_n^{(t)}(k) + m_{n-1}^{(t)}(k)\big) \le 2\rho + 2\bar\rho\,\mathfrak{h}_n.$$

By (2.8) and $2|q_{n-1}q_n| \le m_{n-1}+m_n$, $|\dot a_n| \le \sum_k |\varphi(k)|\big(|x_k - a_n|m_n(k) + b_n(m_{n-1}(k)+m_n(k))\big)$. Let $n \ge n_0$, with n_0 as in Lemma 3.3 and $2M_0^p n_0^{-p} \le 1$. Off the head, $|x_k - a_n| \le 2M_0^p n^{-p}$ and $b_n \le 2M_0^p n^{-p}$, so these indices contribute at most $6M_0^p\rho\, n^{-p}$; on the head use $|x_k - a_n| \le 1$ and $b_n \le 1$, a contribution of at most $3\bar\rho\,\mathfrak{h}_n$. Integrate over $t \in [0,1]$ and apply Lemma 3.3. □

3.3. **Proof of Theorem 1.2.** Let ℓ be slowly varying and $\ell_1 := e^{\Lambda_\ell(\log\cdot)}$. By Lemma 3.2(iii), ℓ_1 is analytic of every angle, so Theorem 2.4 gives $|n^p a_n(\sigma_{p,s,\ell_1}) - 2c| + |n^p b_n(\sigma_{p,s,\ell_1}) - c| = O(1/n)$. Write $\rho := \rho_\ell(\lceil n/M_0\rceil)$ and $\Delta_n := \log b_n^2(\sigma_{p,s,\ell}) - \log b_n^2(\sigma_{p,s,\ell_1})$. By Proposition 3.4, $|\Delta_n| \le 2\rho + Cn^{s+5}e^{-2n}$ for $n \ge n_0$; in particular Δ_n is bounded, since $\rho \le \rho_\ell(1)$, so $|e^{\Delta_n/2} - 1| \le C'|\Delta_n|$ and

$$|n^p b_n(\sigma_{p,s,\ell}) - c| \le |n^p b_n(\sigma_{p,s,\ell_1}) - c| + n^p b_n(\sigma_{p,s,\ell_1})\,|e^{\Delta_n/2} - 1| = O(1/n + \rho + n^{s+5}e^{-2n}).$$

Similarly, the second bound of Proposition 3.4 gives $|n^p a_n(\sigma_{p,s,\ell}) - 2c| \le |n^p a_n(\sigma_{p,s,\ell_1}) - 2c| + 6M_0^p\rho + Cn^{p+s+5}e^{-2n}$. The exponentially small terms are $O(1/n)$, and finitely many n affect only the constant. This is (1.5), and the limits follow because $\rho_\ell(m) \to 0$ (Lemma 3.2(i)). □

Remark 3.5 (the order of the steps). Theorem 2.4 is proved in [9] for every $s > 1$, by an analysis in a window of exponents followed by the Christoffel shift with rate [9, Theorem 3.1]. The smoothing is applied here at the final exponent s, after the shift. Smoothing first and shifting afterwards would also work, but the shift would then have to carry the rate $1/n + \rho_\ell(\lceil n/M_0\rceil)$, which is in general not of the regular form required there.

Remark 3.6 (what the rate measures). For ℓ that is itself analytic of some angle, (1.5) is weaker than Theorem 2.4, since ρ_ℓ need not be $O(1/n)$. Its point is the case of factors that are slowly varying but irregular at every scale, such as $\ell(k) = 1 + \frac{1}{2}(-1)^k/\log(k+1)$, for which $\rho_\ell(m) = O(1/\log m)$ (Section 7).

4. Rough weights

This section proves Theorem 1.3. Throughout, $s > 1$ is fixed, $v_k := k^{-s}$ is the pure power, a superscript v marks quantities of $\sigma_{p,s}$, and $w = ve^{\varphi}$ with φ taking the values $0, \pm\varepsilon$.

By Lemma 2.16, $\log(b_N^2/b_{N-1}^2)$ and a_N move, to first order, by the pairing of φ with $\Delta^2 m_N^v$ and with $\Delta\mathcal{E}_N^v$, up to a remainder of order ε^2 that holds for every base measure. For the pure power these two vectors stay of size $\asymp 1$ (after the natural scaling) on a window of indices of size N. This follows from the arcsine limit of Lemma 2.20: a degenerate vector would force the spectral weights to concentrate near three points (Lemmas 4.1 and 4.3). Choosing $\varphi = \pm\varepsilon$ with the sign of that vector on the window produces a first-order response of size $\asymp \varepsilon$, which the remainder cannot cancel when ε is small. Repeating this on widely separated windows gives infinitely many such responses. The windows do not interact, because the first-order term is evaluated at the pure power, where the other windows lie either in the saturated head (Corollary 2.19) or beyond Mn, where the arcsine limit leaves little mass.

4.1. **Windows and the non-degeneracy of the second difference.** A *window* is a pair $W = (\tau_1, \tau_2)$ with $0 < \tau_1 < \tau_2$; put $W_N := \mathbb{N} \cap [\tau_1 N, \tau_2 N]$. It is *admissible* if $\tau_2 > u_{\mathrm{sat}}(p)$. Let $Z := \{-2, 0, 2\}$, and put

$$\begin{aligned} I_W &:= (\tau_2^{-p}, \tau_1^{-p}) \cap (0, 4c), \qquad \Theta_W := \{\theta \in (0,\pi) :\ 2c(1+\cos\theta) \in I_W\}, \\ \iota_W &:= \frac{1}{\pi}\int_{\Theta_W} \operatorname{dist}(2\cos\theta, Z)^2 \,\mathrm{d}\theta, \qquad \delta_W := \frac{\iota_W}{32}, \qquad J_W := \Big\lceil \frac{2}{\delta_W} \Big\rceil, \qquad \gamma_W := \frac{\delta_W}{J_W^2}. \end{aligned} \tag{4.1}$$

These depend only on p, τ_1, τ_2. The window is admissible if and only if $I_W \neq \emptyset$, since $\tau_2^{-p} < 4c$ is $\tau_2 > u_{\mathrm{sat}}$. Then $\iota_W > 0$: the map $\theta \mapsto 2c(1+\cos\theta)$ takes $(0,\pi)$ onto $(0,4c)$, so Θ_W is a non-empty open set, and the integrand vanishes only at $\theta \in \{0, \pi/2, \pi\}$. By Lemma 2.23, every window with $\tau_2 \geq 1$ is admissible.

Since arc_{4c} is the image of $\mathrm{d}\theta/\pi$ under $y = 2c(1+\cos\theta)$, and $(y-2c)/c = 2\cos\theta$,

$$\iota_W = \int_{I_W} \operatorname{dist}\big((y-2c)/c, Z\big)^2 \,\mathrm{d}\,\mathrm{arc}_{4c}(y). \tag{4.2}$$

Lemma 4.1 (window non-degeneracy)**.** *Let W be admissible. There is N_0 such that every $N \geq N_0$ has some $m \in [N - J_W + 1, N]$ with*

$$\big\|\Delta^2 m^v_m|_{W_N}\big\|_1 \geq \gamma_W.$$

The proof is by contradiction. If the second differences are small on W_N for J_W consecutive indices, then so are the first differences, so $|q_N|$, $|q_{N-1}|$ and $|q_{N-2}|$ nearly agree on W_N. The three-term recurrence then forces the rescaled atoms in W_N, weighted by m_{N-1}, to lie near the three points Z, which contradicts the arcsine limit, whose mass on I_W is spread out.

Proof. Write $S := W_N$, $\delta := \delta_W$, $J := J_W$, $\gamma := \gamma_W$, $D_m := m_m - m_{m-1}$, and drop the superscript v. Suppose that $\|\Delta^2 m_m|_S\|_1 < \gamma$ for every $m \in [N - J + 1, N]$.

Step 1: first differences are small on S. For $m \in [N-J, N]$, $D_N - D_m = \sum_{j=m+1}^{N} \Delta^2 m_j$, so $\|(D_N - D_m)|_S\|_1 \leq (N-m)\gamma$. Also $m_N - m_{N-J} = \sum_{m=N-J+1}^{N} D_m$ has $\|\cdot|_S\|_1 \leq 2$. Hence $J\|D_N|_S\|_1 \leq 2 + \gamma J(J-1)/2$, that is, $\|D_N|_S\|_1 \leq 2/J + (J-1)\gamma/2$, and since $D_{N-1} = D_N - \Delta^2 m_N$, also $\|D_{N-1}|_S\|_1 \leq 2/J + (J+1)\gamma/2$. As $2/J \leq \delta$ and $\gamma = \delta/J^2$, both are at most 2δ.

Step 2: the moduli nearly agree on S. Put $e := |q_{N-2}| - |q_{N-1}|$ and $e' := |q_N| - |q_{N-1}|$. Since $(\alpha-\beta)^2 \leq |\alpha^2 - \beta^2|$ for $\alpha, \beta \geq 0$, $\sum_{k\in S} e_k^2 \leq \|D_{N-1}|_S\|_1 \leq 2\delta$ and $\sum_{k\in S} e_k'^2 \leq \|D_N|_S\|_1 \leq 2\delta$.

Step 3: the recurrence on S. Fix $k \in S$, let $\epsilon_k := \operatorname{sgn}(q_{N-1}(k) q_{N-2}(k))$ (with $\operatorname{sgn} 0 := 1$), and $t_k := x_k - a_{N-1} - b_{N-1}\epsilon_k$. We claim

$$|q_{N-1}(k)| \cdot \big||t_k| - b_N\big| \leq b_{N-1}|e_k| + b_N|e'_k|.$$

If $q_{N-1}(k) = 0$ there is nothing to prove. Otherwise $q_{N-2}(k) = \epsilon_k \operatorname{sgn}(q_{N-1}(k))|q_{N-2}(k)|$, and (2.3) at index $N-1$, multiplied by $\operatorname{sgn}(q_{N-1}(k))$, gives $b_N|q_N(k)| = \big|(x_k - a_{N-1})|q_{N-1}(k)| - b_{N-1}\epsilon_k|q_{N-2}(k)|\big| = \big|t_k|q_{N-1}(k)| - b_{N-1}\epsilon_k e_k\big|$. Hence $\big|b_N|q_N(k)| - |t_k||q_{N-1}(k)|\big| \leq b_{N-1}|e_k|$, and $b_N|q_N(k)| = b_N|q_{N-1}(k)| + b_N e'_k$, which proves the claim. Now $||t_k| - b_N| = \operatorname{dist}(t_k, \{\pm b_N\}) \geq \operatorname{dist}(x_k - a_{N-1}, \tilde{Z})$ with $\tilde{Z} := \{\pm b_{N-1} \pm b_N\}$. Put $\bar{b} := \max(b_N, b_{N-1})$, $y_k := (x_k - a_{N-1})/\bar{b}$ and $Z_N := \tilde{Z}/\bar{b}$. Squaring the claim, using $(\alpha+\beta)^2 \leq 2\alpha^2 + 2\beta^2$ and Step 2, and summing over S:

$$\sum_{k\in S} m_{N-1}(k)\, \operatorname{dist}(y_k, Z_N)^2 \leq 2\big(\|e|_S\|_2^2 + \|e'|_S\|_2^2\big) \leq 8\delta = \iota_W/4. \tag{4.3}$$

Step 4: the arcsine limit contradicts (4.3) for large N. Put $n := N-1$ and $Y_k := n^p x_k$. Then $y_k = \phi_N(Y_k)$ with $\phi_N(Y) := (Y - n^p a_{N-1})/(n^p \bar{b})$. By Theorem 2.5, $n^p a_{N-1} \to 2c$

and $n^p\bar{b} \to c$, so $\phi_N \to \phi$, $\phi(Y) := (Y - 2c)/c$, uniformly on compact sets, and $Z_N \to Z$ in the Hausdorff distance. Since $\operatorname{dist}(\cdot, Z_N)$ is 1-Lipschitz and differs from $\operatorname{dist}(\cdot, Z)$ by at most the Hausdorff distance, $\operatorname{dist}(\phi_N, Z_N)^2 \to \operatorname{dist}(\phi, Z)^2$ uniformly on compact sets. By (4.2) and monotone convergence there is a continuous h with $0 \le h \le 1$, compact support in the open interval I_W, and $\int h \operatorname{dist}(\phi, Z)^2 \,\mathrm{d}\operatorname{arc}_{4c} \ge \iota_W/2$. Now $k \in S$ if and only if Y_k lies in $[(n/(\tau_2 N))^p, (n/(\tau_1 N))^p]$, whose endpoints tend to τ_2^{-p} and τ_1^{-p}; so for large N the support of h lies inside this interval, and $\mathbf{1}_S(k) \ge h(Y_k)$ for all k. Hence the left side of (4.3) is at least $\int G_N \,\mathrm{d}\nu_{n,0}$ with $G_N := h \operatorname{dist}(\phi_N, Z_N)^2$ and $\nu_{n,0}$ as in Lemma 2.20. The functions G_N converge uniformly on $[0,\infty)$ to the bounded continuous function $G := h \operatorname{dist}(\phi, Z)^2$, so $\int G_N \,\mathrm{d}\nu_{n,0} \to \int G \,\mathrm{d}\operatorname{arc}_{4c} \ge \iota_W/2$ by Lemma 2.20. For N large this contradicts (4.3). □

4.2. **The off-diagonal coefficients.** Recall $F_N = \log(b_N^2/b_{N-1}^2)$. For an admissible window W put

$$M_W := \max\Big(1, \big(4c\sin^2(\pi\gamma_W/32)\big)^{-1/p}\Big), \qquad R_W := \max\big(\tau_2/\tau_1,\ \tau_2 M_0,\ M_W/\tau_1\big).$$

Since T_p of (2.11) is non-increasing and $T_p((4c\sin^2(\pi\gamma_W/32))^{-1/p}) = \gamma_W/16$, we have $T_p(M_W) \le \gamma_W/16$.

Theorem 4.2. *Let W be admissible, let N_0 be as in Lemma 4.1, and let $N_1 < N_2 < \cdots$ be integers with $N_1 \ge \max(N_0, 2J_W + 2)$, $\tau_2 N_i < \tau_1 N_{i+1}$, and either* (A) *$N_{i+1}/N_i \to \infty$, or* (B) *$N_{i+1} > R_W N_i$ for every i. For each i pick $m_i \in [N_i - J_W + 1, N_i]$ as in Lemma 4.1. Let $\varepsilon \in (0, \gamma_W/16]$ and*

$$\varepsilon_k := \varepsilon \operatorname{sgn}\big(\Delta^2 m^v_{m_i}(k)\big) \quad (k \in W_{N_i},\ i \ge 1), \qquad \varepsilon_k := 0 \quad \text{otherwise},$$

with $\operatorname{sgn} 0 := 1$. Then $w_k := k^{-s} e^{\varepsilon_k}$ satisfies $\liminf_i F_{m_i}(w) \ge \varepsilon\gamma_W/2$ in case (A)*, and $\liminf_i F_{m_i}(w) \ge \varepsilon\gamma_W/4$ in case* (B).

Proof. The windows W_{N_i} are disjoint, so $\varphi := (\varepsilon_k)$ takes values in $\{0, \pm\varepsilon\}$. Fix i, write $m := m_i$, and let $H_i := \bigcup_{j<i} W_{N_j} \subset [1, \tau_2 N_{i-1}]$ and $T_i := \bigcup_{j>i} W_{N_j} \subset [\tau_1 N_{i+1}, \infty)$ be the other windows.

Linearisation. Lemma 2.16(ii), applied along the direct path $ve^{t\varphi}$, gives $F_m(w) \ge F_m(v) + \langle \varphi, \Delta^2 m^v_m\rangle - 2\varepsilon^2$.

Own window and cross terms. By the choice of the signs, $\langle \varphi, \Delta^2 m^v_m\rangle = \varepsilon \|\Delta^2 m^v_m|_{W_{N_i}}\|_1 + E_i$, where the first term is at least $\varepsilon\gamma_W$ by Lemma 4.1, and the cross term $E_i := \langle \varphi \mathbf{1}_{H_i \cup T_i}, \Delta^2 m^v_m\rangle$ satisfies $|E_i| \le \varepsilon\big(m^v_m + 2m^v_{m-1} + m^v_{m-2}\big)(H_i \cup T_i)$.

Head. Let $m' \in \{m, m-1, m-2\}$; then $N_i - J_W - 1 \le m' \le N_i$, so $m' \ge N_i/2$. For $k \in H_i$, $k/m' \le 2\tau_2 N_{i-1}/N_i$, which tends to 0 in case (A) and is less than $2\tau_2/R_W \le 2/M_0$ in case (B). So, for i large, $H_i \subset [1, 2m'/M_0]$, and Corollary 2.19 (with $c_1 = 1$, $A = s$, $\|v\|_1 = \zeta(s)$) gives $m^v_{m'}(H_i) \le \zeta(s) m'^{s+1}(m'+1)^2 e^{-2m'} \to 0$.

Tail. In case (A) fix $M \ge 1$. For i large, $\tau_1 N_{i+1} > M N_i \ge M m'$, so $T_i \subset (Mm', \infty)$ and, by Lemma 2.20, $\limsup_i m^v_{m'}(T_i) \le T_p(M)$; as M is arbitrary, $m^v_{m'}(T_i) \to 0$. In case (B), $\tau_1 N_{i+1} > \tau_1 R_W N_i \ge M_W m'$, so $\limsup_i m^v_{m'}(T_i) \le T_p(M_W) \le \gamma_W/16$.

Conclusion. Hence $\limsup_i |E_i|$ is 0 in case (A) and at most $4\varepsilon\gamma_W/16$ in case (B). By Theorem 2.5, $b_m(v)/b_{m-1}(v) \to 1$, so $F_m(v) \to 0$. With $\varepsilon \le \gamma_W/16$ we have $2\varepsilon^2 \le \varepsilon\gamma_W/8$, and therefore $\liminf_i F_{m_i}(w)$ is at least $\varepsilon\gamma_W(1 - \frac{1}{8})$ in case (A) and $\varepsilon\gamma_W(1 - \frac{1}{8} - \frac{1}{4})$ in case (B); in particular it is at least $\varepsilon\gamma_W/2$ and $\varepsilon\gamma_W/4$, respectively. □

Proof of Theorem 1.3(i). Integers N_i as in Theorem 4.2 exist (choose them recursively), the weights constructed there are of the stated form, and $\limsup_n F_n(w) \ge \liminf_i F_{m_i}(w) \ge \varepsilon\gamma_W/4$. □

4.3. **The diagonal coefficients.** For the diagonal the relevant vector is $\Delta\mathcal{E}_n$ of Lemma 2.14, and its non-degeneracy has a direct proof, with an explicit constant and a window of two indices. For $z \in \mathbb{R}$ put

$$\Upsilon(z) := \min_{\chi\in[0,\pi]} \big(|\sin 2\chi - z\cos^2\chi| + |z\sin^2\chi - \sin 2\chi|\big),$$

and

$$\mathfrak{i} := \frac{1}{\pi}\int_0^\pi \min\big(1, \Upsilon(2\cos\theta)\big)\,\mathrm{d}\theta. \tag{4.4}$$

The function Υ is 1-Lipschitz, since the two terms change by at most $|z-z'|\cos^2\chi$ and $|z-z'|\sin^2\chi$. It vanishes exactly on $Z = \{-2, 0, 2\}$: with $u = \cos\chi$, $u' = \sin\chi$ the two terms are $|u(2u' - zu)|$ and $|u'(zu' - 2u)|$; if $u = 0$ the second is $|z|$, if $u' = 0$ the first is $|z|$, and if $uu' \neq 0$ both vanish only if $2u' = zu$ and $zu' = 2u$, whence $z^2 = 4$; conversely the minimum is 0 for $z \in Z$, and it is attained. Hence $0 < \mathfrak{i} \le 1$, because the integrand in (4.4) is continuous, non-negative, and vanishes only at $\theta \in \{0, \pi/2, \pi\}$.

Lemma 4.3 (non-degeneracy of the diagonal response)**.** *For the pure power,*

$$\liminf_{N\to\infty}\Big(\big\|N^p\Delta\mathcal{E}^v_N\big\|_1 + \big\|N^p\Delta\mathcal{E}^v_{N+1}\big\|_1\Big) \ge 2c\,\mathfrak{i}.$$

Consequently, with $\gamma_{\mathrm{a}} = c\mathfrak{i}/2$, there is N_0 such that every $N \ge N_0$ has some $n \in \{N, N+1\}$ with $\|n^p\Delta\mathcal{E}^v_n\|_1 \ge \gamma_{\mathrm{a}}$.

Proof. Fix N and write $u := q_N$, $u' := q_{N+1}$, $B := b_{N+1}$ (all for v). By (2.9) at the indices N and $N+1$, at every atom

$$\Delta\mathcal{E}_N = 2Buu' - (x - a_N)u^2, \qquad \Delta\mathcal{E}_{N+1} = (x - a_{N+1})u'^2 - 2Buu'.$$

Put $y_k := N^p x_k$ and $z_k := (y_k - 2c)/c$. Then $N^p\Delta\mathcal{E}_N = c(2uu' - zu^2) + r_N$ and $N^p\Delta\mathcal{E}_{N+1} = c(zu'^2 - 2uu') + r'_N$, where, by $2|uu'| \le u^2 + u'^2$,

$$|r_N| + |r'_N| \le \delta_N(u^2 + u'^2), \qquad \delta_N := 2|N^pB - c| + |N^pa_N - 2c| + |N^pa_{N+1} - 2c|,$$

and $\delta_N \to 0$ by Theorem 2.5. The expression $|2uu' - zu^2| + |zu'^2 - 2uu'|$ is homogeneous of degree 2 in (u, u'), so it is at least $\Upsilon(z)(u^2 + u'^2)$ at every atom. Summing over the atoms, with $\sum_k(u_k^2 + u_k'^2) = 2$,

$$\|N^p\Delta\mathcal{E}_N\|_1 + \|N^p\Delta\mathcal{E}_{N+1}\|_1 \ge c\big\langle \min(1, \Upsilon(z)), m_N + m_{N+1}\big\rangle - 2\delta_N.$$

By Lemma 2.20 with $i = 0$ and $i = 1$, the images of m_N and m_{N+1} under $k \mapsto y_k$ converge weakly to arc_{4c}, and $y \mapsto \min(1, \Upsilon((y-2c)/c))$ is bounded and continuous; substituting $y = 2c(1 + \cos\theta)$, the pairing tends to $2\mathfrak{i}$. This proves the first claim. For $N \ge N_0$ the sum is then at least $c\mathfrak{i}$, so one of the two terms is at least $c\mathfrak{i}/2$; and $\|n^p\Delta\mathcal{E}_n\|_1 \ge \|N^p\Delta\mathcal{E}_n\|_1$ for $n \ge N$. □

Theorem 4.4. *Let $\varepsilon \in (0, \mathfrak{i}/16]$. Let N_0 be as in Lemma 4.3, and choose integers $n_1 < n_2 < \cdots$ with $\|n_i^p\Delta\mathcal{E}^v_{n_i}\|_1 \ge \gamma_{\mathrm{a}}$, $n_1 \ge \max(N_0, 2)$ and $n_{i+1} \ge (i+1)^2M_0n_i$. Put $W_i := \mathbb{N} \cap [n_i/M_0,\, in_i]$,*

$$\varepsilon_k := (-1)^i\varepsilon\,\operatorname{sgn}\big(\Delta\mathcal{E}^v_{n_i}(k)\big) \quad (k \in W_i), \qquad \varepsilon_k := 0 \quad \textit{otherwise},$$

and $w_k := k^{-s}e^{\varepsilon_k}$. Then $(-1)^i\big(n_i^pa_{n_i}(w) - 2c\big) \ge \varepsilon\gamma_{\mathrm{a}}/2 - o(1)$ as $i \to \infty$.

Proof. The indices n_i exist, because every pair $\{N, N+1\}$ with $N \ge N_0$ contains one. The windows are disjoint: the right end in_i of W_i is less than the left end $n_{i+1}/M_0 \ge (i+1)^2n_i$ of W_{i+1}. Fix i and put $A_i := \mathbb{N} \setminus W_i = [1, n_i/M_0) \cup (in_i, \infty)$. Every other window lies in A_i: for $j < i$, $W_j \subset [1, (i-1)n_{i-1}]$ and $(i-1)n_{i-1} \le (i-1)n_i/(i^2M_0) < n_i/M_0$; for $j > i$, $W_j \subset [n_{i+1}/M_0, \infty)$ and $n_{i+1}/M_0 > in_i$.

Off-window mass. By Lemma 2.14 and Theorem 2.5, for n large and every $A \subset \mathbb{N}$,

$$n^p \|\Delta\mathcal{E}_n^v|_A\|_1 \le \tfrac{1}{2} n^p \max(b_n, b_{n+1})(v)\,\big(m_{n-1}^v + 2m_n^v + m_{n+1}^v\big)(A) \le c\big(m_{n-1}^v + 2m_n^v + m_{n+1}^v\big)(A).$$

Let $\omega_i := c(m_{n_i-1}^v + 2m_{n_i}^v + m_{n_i+1}^v)(A_i)$. We claim $\omega_i \to 0$. For $n' \in \{n_i - 1, n_i, n_i + 1\}$, the head $[1, n_i/M_0)$ lies in $[1, 2n'/M_0]$, so Corollary 2.19 makes its $m_{n'}^v$-mass exponentially small. For the tail fix $M \ge 1$: for $i \ge M$, $(in_i, \infty) \subset (Mn_i, \infty)$, and by Lemma 2.20 with the shifts $-1, 0, 1$, $\limsup_i m_{n'}^v((Mn_i, \infty)) \le T_p(M)$. Hence $\limsup_i \omega_i \le 4cT_p(M)$ for every M, and $T_p(M) \to 0$.

Linear response. Lemma 2.16(iii), applied along the direct path $ve^{t\varphi}$ with $\varphi = (\varepsilon_k)$, gives

$$n_i^p \big|a_{n_i}(w) - a_{n_i}(v) - \langle \varphi, \Delta\mathcal{E}_{n_i}^v\rangle\big| \le (e^\varepsilon - 1 - \varepsilon)\, n_i^p \big(b_{n_i} + b_{n_i+1}\big)(v) = (e^\varepsilon - 1 - \varepsilon)(2c + o(1)).$$

By the choice of signs, and because φ vanishes on A_i except on the other windows,

$$(-1)^i n_i^p \langle \varphi, \Delta\mathcal{E}_{n_i}^v\rangle \ge \varepsilon\, n_i^p \|\Delta\mathcal{E}_{n_i}^v|_{W_i}\|_1 - \varepsilon\, n_i^p \|\Delta\mathcal{E}_{n_i}^v|_{A_i}\|_1 \ge \varepsilon(\gamma_{\mathrm{a}} - 2\omega_i).$$

Conclusion. Since $\varepsilon \le \mathfrak{i}/16 \le 1/16$, $e^\varepsilon - 1 - \varepsilon \le \frac{1}{2}\varepsilon^2 e^\varepsilon \le 0.54\,\varepsilon^2$, and so $2c(e^\varepsilon - 1 - \varepsilon) \le 1.08\, c\varepsilon\mathfrak{i}/16 < \varepsilon\gamma_{\mathrm{a}}/2$. With $n^p a_n(v) \to 2c$ (Theorem 2.5) we obtain $(-1)^i(n_i^p a_{n_i}(w) - 2c) \ge \varepsilon\gamma_{\mathrm{a}} - \varepsilon\gamma_{\mathrm{a}}/2 - o(1)$. □

Proof of Theorem 1.3(ii). The weights of Theorem 4.4 have the stated form, and the values at even and odd i give the bounds for the lim sup and the lim inf. □

4.4. **The averaged law, and remarks.**

Proof of Theorem 1.3(iii). If $e^{-\varepsilon}k^{-s} \le w_k \le e^\varepsilon k^{-s}$, then $\sum_k w_k \le e^\varepsilon \zeta(s)$ and $w_k \ge e^{-\varepsilon}k^{-s}$, so Theorem 2.1 gives (1.7) and (1.8). In case (i), suppose that $n^p b_n \to L \in [0, \infty]$. Then $\log(j^p b_j) \to \log L \in [-\infty, \infty]$, and the weights $2(n-j)/n^2$ in (1.8) are non-negative with sum $(n-1)/n \to 1$, so the left side of (1.8) tends to $\log L$; hence $L = c$. But then $\log(b_n^2/b_{n-1}^2) = \log(\beta_n^2/\beta_{n-1}^2) - 2p\log\frac{n}{n-1} \to 0$ with $\beta_n := n^p b_n/c$, which contradicts (i). In case (ii), the lim sup and the lim inf of $n^p a_n$ differ by at least $\varepsilon\gamma_{\mathrm{a}}$. □

Remark 4.5 (size of the constants). The constants of Theorem 1.3 are far from the size of the effect. For $p = 2$ and $W = (0.6, 1.6)$ a quadrature gives $\iota_W \approx 0.2345$, hence $J_W = 273$, $\gamma_W \approx 9.8\cdot 10^{-8}$ and $R_W \approx 1.1\cdot 10^8$; for $p = 1$, $\iota_W \approx 0.127$ and $\gamma_W \approx 1.6\cdot 10^{-8}$. Numerically, for the pure power with $p = s = 2$, the minimum of $\|\Delta^2 m_m|_{W_N}\|_1$ over $m \in [N-7, N]$ is 1.06, 1.09 and 1.09 for $N = 100, 200, 300$, and separation by a factor 4 already suffices (Section 7). The loss comes from the contradiction in Lemma 4.1, which costs the factor J_W^2. For the diagonal, a grid quadrature gives $\mathfrak{i} \approx 0.314$, and the observed first-order response $\varepsilon\|n^p\Delta\mathcal{E}_n^v|_W\|_1$ is about $0.8\,\varepsilon c(p)$ for $p = s = 2$, about ten times the guaranteed $\varepsilon\gamma_{\mathrm{a}}/2$. None of these numerical values is used in the proofs, which use only $\iota_W > 0$ and $\mathfrak{i} > 0$.

Remark 4.6 (the law to accuracy ε^2). Lemma 2.16(i) shows what a perturbation of size ε can do to the off-diagonal law. Let φ be bounded with $\frac{1}{2}\operatorname{osc}\varphi \le \varepsilon$, $w = ve^\varphi$, and $\beta_n := n^p b_n/c$. Then $\log\beta_n^2(w) = \langle \varphi, m_n^v - m_{n-1}^v\rangle + r_n + \log\beta_n^2(v)$ with $|r_n| \le \varepsilon^2$ and $\log\beta_n^2(v) \to 0$, so

$$\Big|\limsup_n |\log\beta_n^2(w)| - \limsup_n |\langle \varphi, m_n^v - m_{n-1}^v\rangle|\Big| \le \varepsilon^2.$$

Replacing φ by $\tau\varphi$ and letting $\tau \downarrow 0$, the defect $\limsup_n |\log\beta_n^2(ve^{\tau\varphi})|$ is $o(\tau)$ if and only if $\langle \varphi, m_n^v - m_{n-1}^v\rangle \to 0$. So a rough perturbation breaks the law at first order exactly when it correlates with $m_n^v - m_{n-1}^v$ along infinitely many n, which is why the construction uses sign patterns tied to the spectral weights. For fixed φ this leaves a band of width ε^2 that we cannot close; in particular we do not know whether $\langle \varphi, m_n^v - m_{n-1}^v\rangle \to 0$ implies the law for ve^φ.

Remark 4.7 (admissibility). The condition $\tau_2 > u_{\mathrm{sat}}$ is necessary for this mechanism. If $\tau_2 \le u_{\mathrm{sat}}$ and $m - N$ stays bounded, the image of W_N under $k \mapsto m^p x_k$ lies in $[(m/(\tau_2 N))^p, \infty)$, whose left end tends to $\tau_2^{-p} \ge 4c$; since arc_{4c} has no mass beyond $4c$ and no atom, Lemma 2.20 gives $m_m^v(W_N) \to 0$, so $\|\Delta^2 m_m^v|_{W_N}\|_1 \to 0$ and the first-order gain disappears. In the variable $u = k/N$ of Proposition 2.2, such windows lie in $(0, u_{\mathrm{sat}}]$, where the equilibrium density ρ^* equals 1; the argument just given shows that they carry asymptotically no spectral weight.

Remark 4.8. The construction of Theorem 4.4 controls a_n only. We do not know whether b_n, or the ratio $a_n/(2b_n)$, converges for the same weights.

5. The Christoffel transformation at the Cesàro level

The direction $\varphi = -p \log k$ is special: $k^{-p} = x_k$, so $e^{-p \log k} v_k = x_k v_k$, and the end point of the path $ve^{t\varphi}$ at $t = 1$ is the measure $x\sigma$. For the pure powers this is the shift $s \mapsto s+p$. Here no first-variation estimate is needed, because the Jacobi matrices of σ and $x\sigma$ are related exactly by the factorisation of Lemma 2.7. This section uses that factorisation beyond the pointwise transfer of [9]: for the converse direction and a point mass at 0 (Proposition 5.1), and for the product identity (Proposition 5.2), which with the chain-limit statement of Lemma 2.7(ii) gives Theorem 1.4(i). Combined with a comparison lemma for perturbations of relative size $O(1/k)$ and two explicit measures, the factorisation determines the averaged $1/n$ coefficient on lattices of exponents (Corollary 5.5). Finally, it relates the pointwise $1/n$ coefficients at s and $s + p$ (Proposition 5.6). In this section σ is a positive Borel measure on $[0, \infty)$ with infinite support and finite moments, u_n, l_n are as in (2.5), and $\tilde{a}_n, \tilde{b}_n$ are the coefficients of $x\sigma$.

5.1. The converse, and a point mass at zero.

Proposition 5.1. *Let $c' > 0$. If $n^p \tilde{a}_n \to 2c'$ and $n^p \tilde{b}_n \to c'$, then $n^p a_n \to 2c'$ and $n^p b_n \to c'$. Consequently, for every $m_0 \ge 0$, the measure $\sigma + m_0\delta_0$ satisfies $n^p a_n \to 2c'$, $n^p b_n \to c'$ if and only if σ does.*

Proof. By Lemma 2.7(i), $l_{n+1} = \tilde{a}_n - u_n > 0$ and $u_{n+1} = \tilde{b}_{n+1}^2 / l_{n+1}$, so $z_n := n^p u_n / c'$ satisfies

$$z_{n+1} = \frac{B_n}{A_n - z_n}, \qquad 0 < z_n < A_n, \qquad A_n := \frac{n^p \tilde{a}_n}{c'} \to 2, \qquad B_n := \frac{(n+1)^p n^p \tilde{b}_{n+1}^2}{c'^2} \to 1.$$

The limiting map $z \mapsto 1/(2-z)$ has the double fixed point 1, and $1/(2-z) \ge z$ for $z < 2$. Fix $\delta \in (0, \frac{1}{5}]$, put $\epsilon := \delta^3/16$, and choose n_5 with $|A_n - 2|, |B_n - 1| \le \epsilon$ for $n \ge n_5$. For $n \ge n_5$, $z_{n+1} < A_{n+1}$ gives $A_n - z_n > B_n/A_{n+1} \ge (1-\epsilon)/(2+\epsilon) > 0.49$; hence $z_n < 1.52$, $2 - z_n > 0.48$, and

$$\left| z_{n+1} - \frac{1}{2 - z_n} \right| \le \frac{|B_n - 1|}{A_n - z_n} + \frac{|A_n - 2|}{(A_n - z_n)(2 - z_n)} \le 7\epsilon.$$

If $z_m \ge 1 + \delta$ for some $m \ge n_5$, then $z_{m+1} \ge z_m + (z_m - 1)^2/(2 - z_m) - 7\epsilon \ge z_m + \delta^2/2$, and by induction z would exceed 1.52; so $z_n < 1+\delta$ for $n \ge n_5$. If $z_n \le 1-\delta$, then $z_{n+1} \ge z_n + \delta^2/4$, so z enters $(1-\delta, 1+\delta)$ after finitely many steps; and if $z_n > 1-\delta$, then $z_{n+1} \ge 1/(1+\delta) - 7\epsilon > 1-\delta$. Hence $z_n \to 1$. Finally $a_n = u_n + l_n = u_n + \tilde{a}_{n-1} - u_{n-1}$ and $b_n^2 = u_{n-1} l_n$ give $n^p a_n \to 2c'$ and $n^{2p} b_n^2 \to c'^2$.

For the second statement, $\sigma + m_0\delta_0$ has infinite support and finite moments, and $x(\sigma + m_0\delta_0) = x\sigma$. By Theorem 2.8 and the first part, the law for either measure is equivalent to the law for $x\sigma$. □

The first statement also follows from the proposition of Dai, Piao and Qiao quoted in the introduction. With $\mathsf{g}_n := u_n/\tilde{a}_n \in (0, 1)$ one has $\tilde{b}_n^2/(\tilde{a}_{n-1}\tilde{a}_n) = (1 - \mathsf{g}_{n-1})\mathsf{g}_n$, so these numbers

form a chain sequence with parameter sequence (g_n). The hypothesis makes them tend to $\frac{1}{4}$, so $\mathsf{g}_n \to \frac{1}{2}$ by [1, Proposition 2.1]; this gives $n^p u_n \to c'$ and $n^p l_{n+1} \to c'$. For discrete measures with positive summable weights and bounded, strictly decreasing, non-negative atoms, the first statement of Proposition 5.1 has been formalised in the Lean 4 proof assistant, with a general positive scale sequence whose consecutive ratios tend to 1 in place of n^p.

5.2. The product identity and Theorem 1.4(i).

Proposition 5.2. *For every $n \geq 1$,*

$$\prod_{m=1}^{n} \Big(\frac{\tilde{b}_m}{b_m}\Big)^2 = \frac{u_n}{u_0}, \qquad u_0 = a_0 = \frac{\int x \,\mathrm{d}\sigma}{\int \mathrm{d}\sigma}.$$

Proof. By Lemma 2.7(i), $b_m^2 = u_{m-1} l_m$ and $\tilde{b}_m^2 = u_m l_m$, so $(\tilde{b}_m/b_m)^2 = u_m/u_{m-1}$; the product telescopes. Also $u_0 = -P_1(0)/P_0(0) = a_0$, since $P_1(x) = x - a_0$. □

Proof of Theorem 1.4(i). Take $\sigma = \sigma_{p,s}$, so that $x\sigma = \sigma_{p,s+p}$ and $u_0 = \zeta(s+p)/\zeta(s)$. Proposition 5.2 gives

$$2\sum_{m=1}^{n} \log \frac{b_m(s+p)}{b_m(s)} = \log\big(n^p u_n\big) - p\log n - \log \frac{\zeta(s+p)}{\zeta(s)}.$$

By Theorem 2.5, $n^p a_n(s) \to 2c$ and $n^p b_n(s) \to c$, so $n^p u_n \to c$ by Lemma 2.7(ii). □

Only the law without rate is used. With the rate $O(1/n)$, the parameters converge only like $n^p u_n - c = O(n^{-1/2})$ in general [9, Lemma 3.5 and Remark 3.6], so the convergence in Theorem 1.4(i) has no better rate a priori; numerically the error is of order $1/n$ (Section 7).

5.3. Perturbations of relative size $O(1/k)$.

Proposition 5.3. *Let v, w be positive weights on the atoms k^{-p} with $\sum_k v_k < \infty$, $v_k \geq c_1 k^{-A}$ for some $c_1 > 0$, $A \geq 0$ and all k, and $|\log(w_k/v_k)| \leq C/k$ for all $k \geq 1$. Then $\log h_n(w) - \log h_n(v) = O(1/n)$, and consequently*

$$\sum_{j=1}^{n} \log \frac{b_j(w)}{b_j(v)} = -\frac{1}{2} \log \frac{\sum_k w_k}{\sum_k v_k} + O(1/n).$$

Proof. Let $\varphi := \log(w/v)$ and $w(t) := ve^{t\varphi}$, $t \in [0,1]$; then $e^{-C}v \leq w(t) \leq e^{C}v$. By (2.6), $\frac{\mathrm{d}}{\mathrm{d}t} \log h_n = \langle \varphi, m_n^{(t)} \rangle$. For $k \geq n/M_0$, $|\varphi(k)| \leq CM_0/n$. The head $[1, n/M_0)$ lies in $[1, 2n/M_0]$, so Corollary 2.19, with $\|w(t)\|_1 \leq e^C \|v\|_1$ and $w_k(t) \geq e^{-C} c_1 k^{-A}$, bounds its $m_n^{(t)}$-mass by $Cn^{A+3}e^{-2n}$, and there $|\varphi| \leq C$. Hence $|\frac{\mathrm{d}}{\mathrm{d}t} \log h_n| \leq CM_0/n + O(n^{A+3}e^{-2n})$ uniformly in t; integrate over $[0,1]$. The second statement follows from $h_n = h_0 \prod_{j \leq n} b_j^2$, where h_0 is the total mass. □

5.4. Two anchors and the lattice values.

The averaged $1/n$ coefficient is fixed by an explicit measure at one exponent; Theorem 1.4(i) then moves it along the lattice $s + p\mathbb{N}_0$. At $p = 1$ the explicit measure is a Tricomi–Carlitz measure, which lives on a symmetric set of atoms; the following standard lemma passes to the squared variable.

Lemma 5.4. *Let ν be a positive measure on $\mathbb{R}$, symmetric under $t \mapsto -t$, with infinite support and finite moments, and let its monic orthogonal polynomials R_N satisfy $tR_N = R_{N+1} + \mathsf{e}_N R_{N-1}$ ($\mathsf{e}_0 := 0$, $\mathsf{e}_N > 0$). Let σ be the image of ν under $t \mapsto t^2$. Then σ is a positive*

measure on $[0,\infty)$ *with infinite support and finite moments, its monic orthogonal polynomials* P_m *satisfy* $R_{2m}(t) = P_m(t^2)$*, and its coefficients are*

$$a_m = \mathsf{e}_{2m} + \mathsf{e}_{2m+1} \quad (m \ge 0), \qquad b_m^2 = \mathsf{e}_{2m-1}\mathsf{e}_{2m} \quad (m \ge 1).$$

Proof. By symmetry $R_N(-t) = (-1)^N R_N(t)$, so $R_{2m}(t) = P_m(t^2)$ for a monic polynomial P_m of degree m, and $\int P_m x^j \,\mathrm{d}\sigma = \int R_{2m} t^{2j} \,\mathrm{d}\nu = 0$ for $j < m$. So the P_m are the monic orthogonal polynomials of σ. Two steps of the recurrence give $t^2 R_{2m} = R_{2m+2} + (\mathsf{e}_{2m} + \mathsf{e}_{2m+1})R_{2m} + \mathsf{e}_{2m-1}\mathsf{e}_{2m}R_{2m-2}$, which is the recurrence of P_m with the stated coefficients. □

Corollary 5.5 (lattice values of the averaged $1/n$ coefficient)**.** (a) *For* $p = 1$ *and every* $s \in \frac{3}{2} + \mathbb{N}_0$ *there is* $C(s)$ *with* $\sum_{j=1}^n \log(2jb_j(s)) = -\frac{s-1}{2}\log n + C(s) + o(1)$.

(b) *For* $p = 2$ *and every* $s \in 2+2\mathbb{N}_0$ *there is* $C(s)$ *with* $\sum_{j=1}^n \log(j^2 b_j(s)/c(2)) = -\frac{s-1}{2}\log n + C(s) + o(1)$.

Proof. (a) Let $\psi^{(1)}$ be the Tricomi–Carlitz measure with parameter $\alpha = 1$: by [11, (1.4)–(1.5)] it is the symmetric measure with mass $(k+1)^{k-1}e^{-k}/k!$ at each of the points $\pm(k+1)^{-1/2}$, $k \ge 0$, and the Tricomi–Carlitz polynomials $f_n = f_n^{(1)}$ are orthogonal with respect to it. By [11, (1.9)] their leading coefficient is $\prod_{k<n}(k+1)/(k+1) = 1$, so they are monic, and [11, (1.6)] reads $(n+1)f_{n+1}(x) = (n+1)xf_n(x) - f_{n-1}(x)$, that is, $\mathsf{e}_N = 1/(N+1)$ for $N \ge 1$. The image of $\psi^{(1)}$ under $t \mapsto t^2$ has mass $2(k+1)^{k-1}e^{-k}/k! = 2e\, v_{k+1}$ at $1/(k+1)$, where $v_j := j^{j-1}e^{-j}/j!$. By Lemma 5.4, the measure $\sigma_{\mathrm{TC}} := \sum_{j\ge1} v_j \delta_{1/j}$ has the coefficients $b_m^{\mathrm{TC}} = (2m(2m+1))^{-1/2}$.

Next compare $\sigma_{1,3/2}$ with σ_{TC}. Put $\varpi(t) := \Gamma(t+1)e^t/(\sqrt{2\pi}\, t^{t+1/2})$, so that $k^{-3/2} = \sqrt{2\pi}\,\varpi(k)v_k$. By Stirling's formula with the remainder bound for real argument [3, (5.11.1) and §5.11(ii)], $\log\Gamma(k) = (k-\frac12)\log k - k + \frac12\log(2\pi) + R(k)$ with $|R(k)| \le 1/(12k)$; since $\log\varpi(k) = \log\Gamma(k) + \log k + k - \frac12\log(2\pi) - (k+\frac12)\log k = R(k)$, we get $|\log\varpi(k)| \le 1/(12k)$ for all $k \ge 1$. Apply Proposition 5.3 with $w_k = k^{-3/2}$ and the weights $\sqrt{2\pi}\, v_k \ge e^{-1/12}k^{-3/2}$, which have the same coefficients as σ_{TC}:

$$\sum_{j=1}^n \log\frac{b_j(3/2)}{b_j^{\mathrm{TC}}} = -\frac12\log\frac{\zeta(3/2)}{\sqrt{2\pi}\, W_{\mathrm{TC}}} + O(1/n), \qquad W_{\mathrm{TC}} := \sum_j v_j.$$

Since $\log(2jb_j^{\mathrm{TC}}) = -\frac12\log(1+\frac{1}{2j}) = -\frac{1}{4j} + O(j^{-2})$, this gives (a) for $s = \frac32$. By Theorem 1.4(i) with $p = 1$, $\sum_{j\le n}\log(b_j(s+1)/b_j(s)) = -\frac12\log n + \mathrm{const} + o(1)$, so the coefficient decreases by $\frac12$ at each step.

(b) By Proposition 2.9, $j^2 b_j(2)/c(2) = 16j^2(4j+1)^{-1}((4j-1)(4j+3))^{-1/2} = 1 - \frac{1}{2j} + O(j^{-2})$, so (b) holds for $s = 2$; Theorem 1.4(i) with $p = 2$ moves it by -1 at each step. □

5.5. **Transfer of the pointwise** $1/n$ **coefficient.** The pointwise $1/n$ term of $n^p b_n(s)$ is Theorem 2.6, proved by Riemann–Hilbert analysis. The product identity alone shows how the coefficients at s and $s+p$ must be related if they exist, and that their existence determines the $1/n$ term of $a_n/(2b_n)$.

Proposition 5.6. *Let* $p > 0$*,* $s > 1$ *and* $\varepsilon > 0$*, and suppose that there are* $d(s), d(s+p) \in \mathbb{R}$ *with*

$$\frac{n^p b_n(s)}{c} = 1 + \frac{d(s)}{n} + O(n^{-1-\varepsilon}), \qquad \frac{n^p b_n(s+p)}{c} = 1 + \frac{d(s+p)}{n} + O(n^{-1-\varepsilon}).$$

Then $d(s+p) = d(s) - \frac{p}{2}$, *and* $n^p u_n(s) = c(1 + O(n^{-\varepsilon'}))$ *with* $\varepsilon' := \min(\varepsilon, 1)$. *If moreover* $\varepsilon > \frac{1}{2}$, *then*

$$\frac{a_n(s)}{2b_n(s)} = 1 - \frac{p}{2n} + O\big(n^{-2\varepsilon'}\big).$$

Proof. By the hypothesis, $\log(b_m(s+p)/b_m(s)) = (d(s+p) - d(s))/m + O(m^{-1-\varepsilon'})$, so the left side of Theorem 1.4(i) is $(d(s+p) - d(s) + \frac{p}{2}) \log n + O(1)$. Its convergence forces $d(s+p) = d(s) - \frac{p}{2}$. Then, by Proposition 5.2, $\log(u_m/u_{m-1}) = 2\log(b_m(s+p)/b_m(s)) = -p/m + O(m^{-1-\varepsilon'})$, so $\log(n^p u_n)$ converges with error $O(n^{-\varepsilon'})$, and its limit is $\log c$ by Lemma 2.7(ii). For the last claim write $X := b_n/u_{n-1}$ (at the exponent s). By Lemma 2.7(i), $a_n = u_n + b_n^2/u_{n-1}$, so

$$\frac{a_n}{2b_n} = \frac{1}{2}\Big(X + \frac{u_n/u_{n-1}}{X}\Big).$$

Here $u_n/u_{n-1} = 1 - p/n + O(n^{-1-\varepsilon'})$, and $X = 1 + O(n^{-\varepsilon'})$, because $u_{n-1} = c(n-1)^{-p}(1 + O(n^{-\varepsilon'}))$ and $b_n = cn^{-p}(1 + O(1/n))$. Hence $a_n/(2b_n) = \frac{1}{2}(X + X^{-1}) - p/(2n) + O(n^{-1-\varepsilon'}) = 1 + (X-1)^2/(2X) - p/(2n) + O(n^{-1-\varepsilon'})$, which is the claim since $2\varepsilon' \le 1 + \varepsilon'$. □

By Theorem 2.6 the hypothesis holds for every $p > 0$, with $d(s) = -(s-1)/2$, and with $\varepsilon = 1$ if $p \ge 1$ and $\varepsilon = p$ if $0 < p < 1$; the conclusion agrees with that theorem. The point of the proposition is that the relation $d(s+p) = d(s) - \frac{p}{2}$, and the $1/n$ term of $a_n/(2b_n)$ when $\varepsilon > \frac{1}{2}$, follow from the Christoffel structure alone once the expansions of $n^p b_n$ at s and $s+p$ exist.

6. Pure powers as s varies

This section proves Theorem 1.4(ii)–(iii) and Corollary 1.5. Moving s is the direction $\varphi = -\log k$, which is unbounded, so we work with the exact derivative in s (Proposition 6.1) and with a two-sided variational bound for $h_n(s)/h_n(s')$ (Lemma 6.2). The bound involves a positive power of $n^p x$ under m_n, which the pointwise law controls, and a negative power. Near $x = 0$ the negative power is controlled, at finitely many fixed exponents, by the tail bound of Lemma 2.22, which rests on Proposition 2.10. Chaining over a grid of exponents gives boundedness (Proposition 6.3); the arcsine limit identifies the limit (Theorem 6.5). This works on compact subsets of the range $\mathcal{S}_p$ of the tail bound. The Christoffel identity of Theorem 1.4(i) then removes the restriction to $\mathcal{S}_p$ for the averaged law (Lemma 6.6).

Throughout, $p > 0$ is fixed, $m_n^{(s)}$ are the spectral weights of $\sigma_{p,s}$, E_s denotes expectation with respect to $m_n^{(s)}$ (with n understood), $Y := n^p x_k = (n/k)^p$, and

$$\Phi_n(s) := \log h_n(s) + s \log n, \qquad \Theta_n(s) := \sum_k \log(k/n)\, m_n^{(s)}(k) \qquad (n \ge 1).$$

By §2.1, $E_s[Y] = n^p a_n(s)$.

6.1. The derivative in s and a variational bound.

Proposition 6.1. *For* $n \ge 0$ *the function* $s \mapsto h_n(s)$ *is real-analytic on* $(1, \infty)$, *and* $\partial_s \log h_n(s) = -\sum_k (\log k)\, m_n^{(s)}(k)$. *Hence* $\Phi_n'(s) = -\Theta_n(s)$ *for* $n \ge 1$.

Proof. The moments of $\sigma_{p,s}$ are $\zeta(s + pj)$, so $h_n = D_{n+1}/D_n$ is real-analytic, and so are the coefficients of $P_n = P_n^{(s)}$. Differentiating $\sum_k k^{-s} P_n(x_k)^2$ term by term is justified because $|P_n(x_k)| \le \max_{[0,1]} |P_n|$ and $\sum_k k^{-s} \log k$ converges locally uniformly; the term containing $\partial_s P_n$ vanishes, since $\partial_s P_n$ has degree less than n. Hence $\partial_s h_n = -\sum_k (\log k) k^{-s} P_n(x_k)^2$. □

Lemma 6.2 (variational bound)**.** *For $n \ge 1$ and $s, s' > 1$, with $E_{s'}$ taking values in $(0,\infty]$,*

$$-\log E_{s'}\big[(k/n)^{s-s'}\big] \le \Phi_n(s) - \Phi_n(s') \le \log E_{s'}\big[(k/n)^{s'-s}\big].$$

Proof. Write $P := P_n^{(s')}$ and $\langle\cdot,\cdot\rangle_s$ for the inner product of $L^2(\sigma_{p,s})$. Upper bound: $h_n(s)$ is the minimum of $\int Q^2\,\mathrm{d}\sigma_{p,s}$ over monic Q of degree n, so $h_n(s) \le \sum_k k^{-s}P(x_k)^2 = h_n(s')E_{s'}[k^{s'-s}]$; add $(s-s')\log n$ to the logarithms. Lower bound: assume $E_{s'}[k^{s-s'}] < \infty$ and put $\phi(x_k) := k^{s-s'}P(x_k)$. Then $\|\phi\|_s^2 = h_n(s')E_{s'}[k^{s-s'}]$ and $\langle Q,\phi\rangle_s = \langle Q,P\rangle_{s'}$ for every polynomial Q, so ϕ is orthogonal in $L^2(\sigma_{p,s})$ to all polynomials of degree less than n, and $\langle x^n,\phi\rangle_s = h_n(s')$. Writing $x^n = P_n^{(s)} + r$ with $\deg r < n$ and using the Cauchy–Schwarz inequality, $h_n(s') \le h_n(s)^{1/2}\|\phi\|_s$, that is, $h_n(s) \ge h_n(s')/E_{s'}[k^{s-s'}]$. □

In terms of Y, $(k/n)^{s'-s} = Y^{(s-s')/p}$. For $s \ge s'$, the upper bound involves a positive power of Y, the lower bound a negative one.

6.2. Boundedness and the limit on compact subsets of $\mathcal{S}_p$.

Proposition 6.3. *For every compact interval $K = [\underline{s},\bar{s}] \subset \mathcal{S}_p$, $\sup_{n\ge1}\sup_{s,s'\in K}|\Phi_n(s) - \Phi_n(s')| < \infty$.*

Proof. Put $\tau_0 := \frac12\vartheta_*(p,\underline{s})$, with ϑ_* as in Lemma 2.22. Since ϑ_* increases with s, $\tau_0 < \vartheta_*(p,s')$ for every $s' \in K$, and $\tau_0 \le p/4$. Choose a grid $\underline{s} = t_0 < t_1 < \cdots < t_J = \bar{s}$ with $t_{i+1} - t_i \le \tau_0$. For each of the finitely many $t = t_i$, $A_t := \sup_n n^p a_n(t) < \infty$ by Theorem 2.5, and $B_t := \sup_n E_t[(k/n)^{\tau_0}] < \infty$. For $n \ge n_0(t)$, Lemma 2.22, with exponent $\vartheta_0 := \vartheta_*(p,t)$, and Tonelli's theorem give

$$E_t[(k/n)^{\tau_0}] \le 1 + \sum_{k>n} m_n^{(t)}(k)\Big(1 + \int_1^{k/n} \tau_0 M^{\tau_0-1}\,\mathrm{d}M\Big) \le 2 + \int_1^\infty \tau_0 M^{\tau_0-1} C M^{-\vartheta_0}\,\mathrm{d}M < \infty,$$

and for the finitely many $n < n_0(t)$ each expectation is finite because $m_n^{(t)}(k) \le Ck^{-t}$ and $\tau_0 < t - 1$.

Let $s \in [t_i, t_{i+1}]$, $t := t_i$ and $r := (s-t)/p \in [0, \frac14]$. By Lemma 6.2 and Jensen's inequality for the concave function $y \mapsto y^r$, $\Phi_n(s) - \Phi_n(t) \le \log E_t[Y^r] \le r\log E_t[Y] \le \frac14\log^+ A_t$. By the lower bound and $y^{pr} \le 1 + y^{\tau_0}$ for $y > 0$ and $0 \le pr \le \tau_0$, $\Phi_n(s) - \Phi_n(t) \ge -\log(1 + B_t)$. So $|\Phi_n(s) - \Phi_n(t_i)|$ is bounded on each cell uniformly in n, and chaining over the J cells bounds $|\Phi_n(s) - \Phi_n(\underline{s})|$ on K. □

Lemma 6.4. *Let $t \in \mathcal{S}_p$ and let g be continuous on $(0,\infty)$ with $|g(y)| \le C(y^{-a} + y^b)$, where $a \ge 0$, $pa < \vartheta_*(p,t)$ and $0 \le b < 2$. Then $E_t[g(Y)] \to \int g\,\mathrm{d}\,\mathrm{arc}_{4c}$.*

Proof. Choose $a' > a$ with $pa' < \vartheta_*(p,t)$, and put

$$V_- := \sup_{n\ge n_0} E_t[Y^{-a'}], \qquad V_+ := \sup_n E_t[Y^2] = \sup_n n^{2p}\big(a_n^2 + b_n^2 + b_{n+1}^2\big)(t).$$

As in the proof of Proposition 6.3, $V_- < \infty$; and $V_+ < \infty$ by Theorem 2.5, since $E_t[x^2] = \langle x^2 q_n, q_n\rangle$. The arcsine law satisfies the same two bounds, because its density is of order $y^{-1/2}$ at 0, $a' < \frac12$ (as $\vartheta_* < p/2$), and its support is compact. For $0 < \epsilon < 1 < R$ the function $g_{\epsilon,R}(y) := g(\min(\max(y,\epsilon),R))$ is bounded and continuous on $[0,\infty)$, and arc_{4c} has no atom at 0, so Lemma 2.20 gives $E_t[g_{\epsilon,R}(Y)] \to \int g_{\epsilon,R}\,\mathrm{d}\,\mathrm{arc}_{4c}$. For each of these measures,

$$E|g - g_{\epsilon,R}| \le E\big[(|g(Y)| + |g(\epsilon)|)\mathbf{1}_{Y<\epsilon}\big] + E\big[(|g(Y)| + |g(R)|)\mathbf{1}_{Y>R}\big] \le 6CV_-\epsilon^{a'-a} + 6CV_+R^{b-2},$$

since Y^{-a}, ϵ^{-a} and 1 are at most $Y^{-a'}\epsilon^{a'-a}$ on $\{Y < \epsilon\}$, and similarly on $\{Y > R\}$. This tends to 0 uniformly in n as $\epsilon \to 0$ and $R \to \infty$. □

Let $\phi(r) := \log \int y^r \,\mathrm{d}\,\mathrm{arc}_{4c}(y)$ for $r > -\frac{1}{2}$. It is finite, convex and C^2 on $[-\frac{1}{4}, \frac{1}{4}]$, with $\phi(0) = 0$, and there is A_0 with $|\phi(r) - r\phi'(0)| \le A_0 r^2$ on $[-\frac{1}{4}, \frac{1}{4}]$. Moreover

$$\phi'(0) = \int \log y \,\mathrm{d}\,\mathrm{arc}_{4c}(y) = \log(2c) + \frac{1}{\pi}\int_0^\pi \log(1+\cos\theta)\,\mathrm{d}\theta = \log c, \tag{6.1}$$

because $|1+e^{i\theta}|^2 = 2(1+\cos\theta)$ and $\frac{1}{2\pi}\int_0^{2\pi} \log|1+e^{i\theta}|\,\mathrm{d}\theta = 0$. The last identity holds with $re^{i\theta}$, $r<1$, in place of $e^{i\theta}$ by the mean value property of the harmonic function $\log|1+z|$, and passes to $r=1$ by dominated convergence, since $\log 2 \ge \log|1+re^{i\theta}| \ge \log|1+e^{i\theta}| - \frac{1}{2}\log 2$ for $\frac{1}{2} \le r < 1$ (use $|1+re^{i\theta}|^2 \ge r|1+e^{i\theta}|^2$).

Theorem 6.5. *For every compact $K \subset \mathcal{S}_p$,*

$$\sup_{s,s_0\in K}\Big|\Phi_n(s) - \Phi_n(s_0) - \frac{s-s_0}{p}\log c\Big| \longrightarrow 0 \qquad (n\to\infty).$$

Proof. We may assume $K = [\underline{s}, \bar{s}]$. Let τ_0 be as in the proof of Proposition 6.3, let $0 < \tau \le \tau_0$, $r_\tau := \tau/p \le \frac{1}{4}$, and take the grid $t_i := \underline{s} + i\tau$ in K (the last cell may be shorter). For a cell $[t_i, t_{i+1}]$ put $\phi_n(r) := \log E_{t_i}[Y^r]$; it is convex by Hölder's inequality, $\phi_n(0) = 0$, and it is finite for $|r| \le r_\tau$. For s in the cell and $r := (s-t_i)/p \in [0, r_\tau]$, Lemma 6.2 and convexity give

$$-\frac{r}{r_\tau}\phi_n(-r_\tau) \le -\phi_n(-r) \le \Phi_n(s) - \Phi_n(t_i) \le \phi_n(r) \le \frac{r}{r_\tau}\phi_n(r_\tau).$$

By Lemma 6.4 with $g(y) = y^{\pm r_\tau}$ (here $pr_\tau = \tau < \vartheta_*(p,t_i)$ and $r_\tau < 2$), $\phi_n(\pm r_\tau) \to \phi(\pm r_\tau) = \pm r_\tau\phi'(0) + e_\pm$ with $|e_\pm| \le A_0 r_\tau^2$. Hence $|\Phi_n(s) - \Phi_n(t_i) - r\phi'(0)| \le (r/r_\tau)(A_0 r_\tau^2 + \eta_{n,i})$ with $\eta_{n,i} \to 0$, uniformly for s in the cell. Summing over the finitely many cells between $\underline{s}$ and s,

$$\limsup_{n\to\infty}\sup_{s\in K}\Big|\Phi_n(s) - \Phi_n(\underline{s}) - \frac{s-\underline{s}}{p}\phi'(0)\Big| \le A_0 r_\tau \frac{\bar{s}-\underline{s}}{p}.$$

As $\tau \in (0, \tau_0]$ is arbitrary, the left side is 0, and $\phi'(0) = \log c$ by (6.1). Differences of two points of K follow. □

The grid has to be refined after $n\to\infty$; a fixed grid gives only the boundedness of Proposition 6.3. At $s - s_0 = p$ the limit is that of Theorem 1.4(i), since $\Phi_n(s+p) - \Phi_n(s) = \log(n^p u_n(s))$ by Proposition 5.2.

6.3. Removing the restriction to $\mathcal{S}_p$.

Lemma 6.6. *For all $s_0, s_1 > 1$, $\Phi_n(s_1) - \Phi_n(s_0) \to \frac{s_1-s_0}{p}\log c$.*

Proof. For $s > 1$ and $j \ge 0$, Proposition 5.2 and $h_n(s) = \zeta(s)\prod_{m\le n} b_m(s)^2$ give $\Phi_n(s+(j+1)p) - \Phi_n(s+jp) = \log(n^p u_n(s+jp))$, which tends to $\log c$ by the proof of Theorem 1.4(i). Write $s_i = t_i + j_i p$ with $t_i \in (1, 1+p]$ and integers $j_i \ge 0$ $(i = 0, 1)$. Then $\Phi_n(s_i) - \Phi_n(t_i) \to j_i \log c$. Since $(1, 1+p] \subset \mathcal{S}_p$, Theorem 6.5 on the compact set $\{t_0, t_1\}$ gives $\Phi_n(t_1) - \Phi_n(t_0) \to \frac{t_1-t_0}{p}\log c$. Adding, $\Phi_n(s_1) - \Phi_n(s_0) \to (\frac{t_1-t_0}{p} + j_1 - j_0)\log c = \frac{s_1-s_0}{p}\log c$. □

The lemma gives no uniformity outside $\mathcal{S}_p$: a uniform version would need the chain-limit statement of Lemma 2.7(ii) uniformly in s, which we have not proved.

6.4. **Proofs of Theorem 1.4(ii)–(iii) and Corollary 1.5.**

Proof of Theorem 1.4(ii). Since $h_n(s) = \zeta(s)\prod_{j\le n} b_j(s)^2$,

$$2\sum_{j=1}^{n} \log \frac{b_j(s_1)}{b_j(s_0)} = \Phi_n(s_1) - \Phi_n(s_0) - (s_1 - s_0)\log n - \log\frac{\zeta(s_1)}{\zeta(s_0)}.$$

Lemma 6.6 gives the limit for all $s_0, s_1 > 1$, and Theorem 6.5 gives uniformity on compact subsets of $\mathcal{S}_p$. The form with $h_n(s_1)/h_n(s_0)$ is the same statement. □

Proof of Corollary 1.5. Let $p = 1$, so $c = \frac{1}{2}$ and $\log(2jb_j(s)) = \log(j^p b_j(s)/c)$. By Corollary 5.5(a) at $s_0 = \frac{3}{2}$, $\sum_{j\le n}\log(2jb_j(\frac{3}{2})) = -\frac{1}{4}\log n + C(\frac{3}{2}) + o(1)$, and by Theorem 1.4(ii), $\sum_{j\le n}\log(b_j(s)/b_j(\frac{3}{2})) = -\frac{s-3/2}{2}\log n + \text{const} + o(1)$. Adding gives the first formula. The moments of $\sigma_{1,s}$ are $\zeta(s+j)$, so $D_n(s)$ is its n-th Hankel determinant and $D_{n+1}(s)/D_n(s) = h_n(s) = \zeta(s)\prod_{j\le n} b_j(s)^2$. Since $2\sum_{j\le n}\log b_j = 2\sum_{j\le n}\log(2jb_j) - 2\log n! - 2n\log 2$, the second formula follows. □

Proof of Theorem 1.4(iii). Let $K \subset \mathcal{S}_p$ be compact, fix $s_0 \in K$ and put $\Delta_n(s) := \Phi_n(s) - \Phi_n(s_0)$. For $n \ge 2$, $\log b_n(s)^2 = \log h_n(s) - \log h_{n-1}(s)$, so

$$\log\frac{b_n(s)^2}{b_n(s_0)^2} = \Delta_n(s) - \Delta_{n-1}(s) - (s - s_0)\log\frac{n}{n-1}.$$

By Theorem 6.5, Δ_n and Δ_{n-1} have the same uniform limit on K, so $b_n(s)/b_n(s_0) \to 1$ uniformly on K; with $n^p b_n(s_0) \to c$ this is the first claim.

For the diagonal, Proposition 5.2 gives $n^p u_n(s) = \exp(\Phi_n(s+p) - \Phi_n(s))$ whenever $s, s+p > 1$, and $a_n = u_n + b_n^2/u_{n-1}$ (Lemma 2.7(i)). If $p > 1$, then $K \cup (K+p)$ lies in a compact subinterval of $\mathcal{S}_p = (1,\infty)$, so $n^p u_n(s) \to c$ uniformly on K by Theorem 6.5, and each of u_n and b_n^2/u_{n-1} is uniformly $cn^{-p}(1+o(1))$ on K. If $0 < p \le 1$ and $1+p \notin K$, write $K = K_1 \cup K_2$ with the compact sets $K_1 := K \cap (1, 1+p)$ and $K_2 := K \cap (1+p, 1+2p)$. On K_1 the previous argument applies, since $K_1 \cup (K_1 + p)$ lies in a compact subinterval of $(1, 1+2p)$. On K_2 use $\sigma_{p,s} = x\,\sigma_{p,s-p}$ and Lemma 2.7(i) for $\sigma_{p,s-p}$:

$$a_n(s) = u_n(s-p) + l_{n+1}(s-p) = u_n(s-p) + \frac{b_{n+1}(s-p)^2}{u_n(s-p)}.$$

Here $(K_2 - p) \cup K_2$ lies in a compact subinterval of $(1, 1+2p)$, so $n^p u_n(s-p) = \exp(\Phi_n(s) - \Phi_n(s-p)) \to c$ uniformly on K_2 by Theorem 6.5, and $(n+1)^p b_{n+1}(s-p) \to c$ uniformly on K_2 by the first claim applied to $K_2 - p$. Hence $n^p a_n(s) \to 2c$ uniformly on K_2. □

At $s = 1+p$ with $p \le 1$ both arguments need Φ_n at an end point of the window ($s+p = 1+2p$ or $s-p = 1$). The pointwise law holds there (Theorem 2.5), but we do not know whether the diagonal law is uniform near $s = 1+p$.

Theorem 1.4(iii) gives no rate. Numerically the deviations are of order $1/n$ (Section 7).

7. Numerical illustrations

The computations below are floating-point illustrations; no proof uses them. The scripts and their stored outputs are in the repository named in the data availability statement. Coefficients were computed by the Lanczos process with full reorthogonalisation in double precision on the atoms $k \le K$, except where a tail correction is mentioned. In that case the atoms $k \le K$ were supplemented by a Gauss–Jacobi rule for the continuous tail $\int_{K+1/2}^{\infty} t^{-s} g(t^{-p})\,\mathrm{d}t$, and the coefficients were computed in quadruple precision by an algorithm that adds one atom at a

time and restores tridiagonal form by plane rotations. This algorithm reproduces the Lommel coefficients of Proposition 2.9 to a relative error below 10^{-14} for $n \le 400$.

Rough weights. Table 1 shows the construction of Theorem 4.2 for $p = s = 2$, the window $W = (0.6, 1.6)$, three scales $N_i = 15, 60, 240$ (separation by the factor 4, far below R_W) and two values of ε, both far above the proved range $\varepsilon \le \gamma_W/16 \approx 6 \cdot 10^{-9}$. At each scale the jump $F_m(w) - F_m(v)$ at the selected index m agrees with the first-order prediction $\langle \varphi, \Delta^2 m^v_m \rangle$ to within 2%, and the contribution of the other windows is at most $1.4 \cdot 10^{-3}$. Every jump exceeds, by a factor larger than 2, the largest $|F_n(w) - F_n(v)|$ over the indices $n \in [10, 246]$ with $|n - m_i| > 2$ for all i. The measure was truncated at $K = 9600$ atoms; with two scales $(30, 240)$ and $\varepsilon = 0.2$, doubling K changes the jumps by less than 10^{-3}.

ε	N_i	m_i	$F_{m_i}(w) - F_{m_i}(v)$	$\langle \varphi, \Delta^2 m^v_{m_i} \rangle$	other windows
0.05	15	15	0.0689	0.0691	$-3.5 \cdot 10^{-4}$
0.05	60	58	0.0624	0.0626	$2.0 \cdot 10^{-4}$
0.05	240	238	0.0601	0.0601	$< 10^{-16}$
0.2	15	15	0.2739	0.2765	$-1.4 \cdot 10^{-3}$
0.2	60	58	0.2457	0.2503	$8.2 \cdot 10^{-4}$
0.2	240	238	0.2389	0.2403	$< 10^{-16}$

TABLE 1. The construction of Theorem 4.2 for $p = s = 2$, $W = (0.6, 1.6)$, scales $15, 60, 240$. The largest $|F_n(w) - F_n(v)|$ over $n \in [10, 246]$ with $|n - m_i| > 2$ is 0.026 for $\varepsilon = 0.05$ and 0.103 for $\varepsilon = 0.2$. Double precision, $K = 9600$ atoms.

The Christoffel–Cesàro identity and the s-law. Table 2 lists the error $S_n := \sum_{m \le n} \log(b_m(s+p)/b_m(s)) + \frac{p}{2} \log n - \frac{1}{2} \log(c\zeta(s)/\zeta(s+p))$ in Theorem 1.4(i), computed with the tail correction ($K = 3.2 \cdot 10^5$ atoms, and $K = 6.4 \cdot 10^5$ for $p = \frac{1}{2}$). It decays like $1/n$, much faster than the a priori rate $n^{-1/2}$. For the uniform statements, on the grid $s \in \{1.5, 1.75, \dots, 4\}$ and with $K = 4 \cdot 10^4$ plus the tail correction, the quantity $\sup_s |\Phi_n(s) - \Phi_n(\frac{3}{2}) - (s - \frac{3}{2}) \log c/p|$ of Theorem 6.5 equals $2.25 \cdot 10^{-2}$, $1.12 \cdot 10^{-2}$, $5.62 \cdot 10^{-3}$, $2.81 \cdot 10^{-3}$ for $p = 1$ and $5.26 \cdot 10^{-2}$, $2.64 \cdot 10^{-2}$, $1.32 \cdot 10^{-2}$, $6.63 \cdot 10^{-3}$ for $p = 2$, at $n = 50, 100, 200, 400$. The deviations $\sup_s |n^p b_n(s)/c - 1|$ and $\sup_s |n^p a_n(s)/(2c) - 1|$ are $3.7 \cdot 10^{-3}$ and $5.0 \cdot 10^{-3}$ ($p = 1$), and $3.7 \cdot 10^{-3}$ and $6.2 \cdot 10^{-3}$ ($p = 2$), at $n = 400$, and halve with each doubling of n. For $p = 1$ the grid extends beyond $\mathcal{S}_p = (1, 3)$; there the computation illustrates what we expect but have not proved. Finally, for the Tricomi–Carlitz comparison in the proof of Corollary 5.5(a), the error of $\sum_{j \le n} \log(b_j(\frac{3}{2})/b_j^{\mathrm{TC}})$, computed with the value $W_{\mathrm{TC}} = 1$ that the computation indicates, is $4.1 \cdot 10^{-4}$ at $n = 100$ and $2.6 \cdot 10^{-5}$ at $n = 1600$.

p	s	$n = 100$	$n = 400$	$n = 800$	$n = 1599$
0.5	1.5	$-6.4 \cdot 10^{-5}$	$-6.7 \cdot 10^{-6}$	$-2.2 \cdot 10^{-6}$	$-7.6 \cdot 10^{-7}$
1	2	$-2.5 \cdot 10^{-3}$	$-6.3 \cdot 10^{-4}$	$-3.1 \cdot 10^{-4}$	$-1.6 \cdot 10^{-4}$
1	3	$2.3 \cdot 10^{-3}$	$6.1 \cdot 10^{-4}$	$3.1 \cdot 10^{-4}$	$1.6 \cdot 10^{-4}$
1.5	1.5	$-8.3 \cdot 10^{-3}$	$-2.1 \cdot 10^{-3}$	$-1.0 \cdot 10^{-3}$	$-5.2 \cdot 10^{-4}$
2	2	$-9.9 \cdot 10^{-3}$	$-2.5 \cdot 10^{-3}$	$-1.2 \cdot 10^{-3}$	$-6.3 \cdot 10^{-4}$

TABLE 2. The error S_n in the Christoffel–Cesàro identity, Theorem 1.4(i). Quadruple precision, atoms plus a quadrature of the continuous tail.

Slowly varying factors. For $p = s = 2$ and the unbounded factor $\ell(k) = \exp\sqrt{\log(k + e)}$, the relative deviation $n^p b_n(\sigma_{p,s,\ell})/(n^p b_n(\sigma_{p,s})) - 1$ equals $1.09 \cdot 10^{-3}$, $5.23 \cdot 10^{-4}$, $2.49 \cdot 10^{-4}$,

$1.18 \cdot 10^{-4}$, $5.6 \cdot 10^{-5}$ at $n = 100, 200, 400, 800, 1600$ (extended precision, $K = 1.2 \cdot 10^5$ and $2.4 \cdot 10^5$ atoms without tail correction, which agree to $2 \cdot 10^{-7}$). The product of n and this deviation decreases slowly, like $\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}\xi}\sqrt{\log(e^\xi + e)}$ at $\xi = \log n$. For the factor $\ell(k) = 1 + \frac{1}{2}(-1)^k/\log(k+1)$ of Remark 3.6, which is slowly varying but rough at every scale, the deviation $|b_n(\sigma_{2,2,\ell})/b_n(\sigma_{2,2}) - 1|$ is $3.9 \cdot 10^{-3}$, $1.5 \cdot 10^{-3}$ and $8.2 \cdot 10^{-4}$ at $n = 250, 1000, 3600$ (extended precision, a single truncation at $K = 1.2 \cdot 10^5$ atoms, so these values are only indicative); the decay is slow, as the rate $\rho_\ell(\lceil n/M_0 \rceil) = O(1/\log n)$ of Theorem 1.2 allows.

8. Open problems

This section records the questions that Theorems 1.2–1.4 leave open, and states two conjectures; the second is taken from [9]. The evidence for them is numerical.

Between slow variation and boundedness. Theorems 1.2 and 1.3 leave a range of regularity conditions open. A simple test case is $w_k = k^{-s}e^{\varepsilon \sin\log k}$: the increments of $\varepsilon \sin\log k$ tend to 0, but $e^{\varepsilon \sin\log k}$ is not slowly varying, so Theorem 1.2 does not apply, and the first-order response of Remark 4.6 has not been analysed for it. For bounded rough weights one may also ask whether the pointwise law holds along a subsequence of density one; we do not know. Averaged forms of the diagonal law, such as Cesàro means of a_n over tails, are the natural counterpart of (1.8) for the diagonal; we do not treat them here.

Uniformity in s. For $0 < p \le 1$, Theorem 1.4(ii)–(iii) are uniform only on compact subsets of $(1, 1 + 2p)$, and the diagonal part of (iii) only away from $s = 1 + p$. Both restrictions come from the range of the tail bound, Proposition 2.10: they would disappear with a version of it beyond $\mathcal{S}_p$, and the first one also with a version of the chain-limit statement of Lemma 2.7(ii) that is uniform in s. A related question is the size of the logarithmic moment $\Theta_n(s) = -\Phi_n'(s)$. Theorem 6.5 shows that its integrals over intervals of s converge, and for each fixed $s \in \mathcal{S}_p$ the bound of Lemma 2.22 (for $k > n$) and the law $n^p a_n \to 2c$ (for $k < n$, via $\log(n/k) \le (n/k)^p/p$) give $\Theta_n(s) = O(1)$; whether $\sup_{s\in K} |\Theta_n(s)| = O(1)$ on compact sets is open. Such a uniform bound would follow from the next conjecture: if $\Phi_n'' = -\Theta_n' \ge 0$ on a compact interval $K' \subset \mathcal{S}_p$ whose interior contains K, then Θ_n is non-increasing on K', so on K it lies between its averages over short intervals of K', and these averages converge by Theorem 6.5.

Conjecture 8.1 (convexity in s)**.** *For every $p > 0$ and every compact $K \subset (1, \infty)$ there is n_0 such that $s \mapsto \log h_n(\sigma_{p,s})$ is convex on K for $n \ge n_0$.*

The evidence is numerical: the second differences in s of $\log h_n$ are positive, from exact determinants for $n \in \{1, 2, 4, 8\}$, $p \in \{\frac{1}{2}, 1, 2\}$ and $s \in \{\frac{3}{2}, 2, \dots, 4\}$, and from the computed coefficients for $n \le 1600$ on the grid $s \in \{\frac{3}{2}, 2, 3, 4\}$, $p \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, 3, 4\}$; at $n = 1600$ the second divided differences are close to $1/(2p^2 n)$, that is, $\partial_s^2 \log h_n \approx 1/(p^2 n)$. So the conjecture concerns the $1/n$ level and is not a soft statement. It also requires more than the exponential-family structure $\sigma_{p,s} = x^{(s-s')/p}\sigma_{p,s'}$, as the following remark shows.

Remark 8.2. Let $f_{ij} := \int p_i p_j \log k \, \mathrm{d}\sigma$, where p_j are the orthonormal polynomials of a measure $\sigma = \sum_k k^{-s}\delta_{k^{-p}}$ with finitely or infinitely many atoms and $\log k$ is evaluated at the atom k^{-p}. Differentiating $\int p_n p_i \, \mathrm{d}\sigma = \delta_{ni}$ in s as in Lemma 2.13 gives $\partial_s^2 \log h_n = \mathrm{Var}_{m_n}(\log k) - 2\sum_{i<n} f_{in}^2$. For the truncation $\sigma^{(N)} := \sum_{k\le N} k^{-s}\delta_{k^{-p}}$ and $n = N-1$, the space $L^2(\sigma^{(N)})$ is spanned by $p_0, \dots, p_{N-1}$, so $\mathrm{Var}_{m_{N-1}}(\log k) = \sum_{i<N-1} f_{i,N-1}^2$ and $\partial_s^2 \log h_{N-1} = -\sum_{i<N-1} f_{i,N-1}^2$. This is negative: otherwise $(\log k)p_{N-1}$ would be a multiple of p_{N-1} on the atoms, so p_{N-1} would vanish at all atoms but one, which is impossible since it is orthogonal to the constants. So convexity fails at the top level of every finite truncation.

The pointwise $1/n$ term. For pure powers the $1/n$ term is known for every $p > 0$ (Theorem 2.6): the $1/n$ coefficients do not depend on whether $p < 1$ or $p \geq 1$, and for $0 < p < 1$ a further term $\mathsf{k}_p n^{-1-p}$ appears in both $n^p b_n/c(p)$ and $n^p a_n/(2c(p))$, so that it cancels in $a_n/(2b_n)$. Corollary 1.5 and Theorem 1.4(ii) give the averaged form of these statements without Riemann–Hilbert analysis.

Slowly varying factors at order $1/n$. The numbers in Section 7 suggest that a slowly varying factor acts at order $1/n$ like a change of s by its local logarithmic derivative. This is the pointwise analogue of Theorem 1.4(ii) along the path $k^{-s}\ell(k)^t$.

Conjecture 8.3 ([9, Conjecture 13.2])**.** *Let $p > 0$, $s > 1$, and let ℓ be analytic of angle ϑ for some $\vartheta > 0$, with $\ell(k) = \exp\Lambda(\log k)$. Then*

$$\frac{n^p b_n(\sigma_{p,s,\ell})}{c(p)} = 1 - \frac{s - 1 - \Lambda'(\log n)}{2n} + o\Big(\frac{\Lambda'(\log n)}{n}\Big) \qquad \text{whenever} \quad n^{\min(1,p)}\Lambda'(\log n) \to \infty.$$

For $0 < p < 1$ the condition $n\Lambda'(\log n) \to \infty$ would not suffice: for $\ell(k) = 1 - \epsilon k^{-p}$, $0 < \epsilon < 1$, the measure $\sigma_{p,s,\ell} = (1 - \epsilon x)\sigma_{p,s}$ is a polynomial Christoffel transform of $\sigma_{p,s}$, and its coefficient of n^{-1-p} is $\mathsf{k}_p + p\, c(p)\epsilon$, not $\frac{1}{2}p\epsilon$ [9, §13].

Other atoms. All results here use the exact atoms k^{-p}: Lemma 2.20 needs a pointwise law for a reference measure, and Section 5 needs $x\sigma_{p,s} = \sigma_{p,s+p}$. For atoms $x_k = f(k)$ regularly varying of index $-p$ the averaged law is [8, Theorem 1.1], and the pointwise law is open.

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Data and code availability

The Lean sources mentioned in Section 5 (declarations `christoffel_shift_law_converse` and `chain_limit`) and the scripts that produced the numbers of Section 7, with their outputs, are available in the repository
https://github.com/speagleg/jacobi-asymptotics under the tag `v1.0-paper3`.

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